TOPIC 8

Development

Topics

This topic looks at:

- the ways in which human development can be measured
- the reasons for differences in levels of development
- what sustainable development is and how it can be achieved.

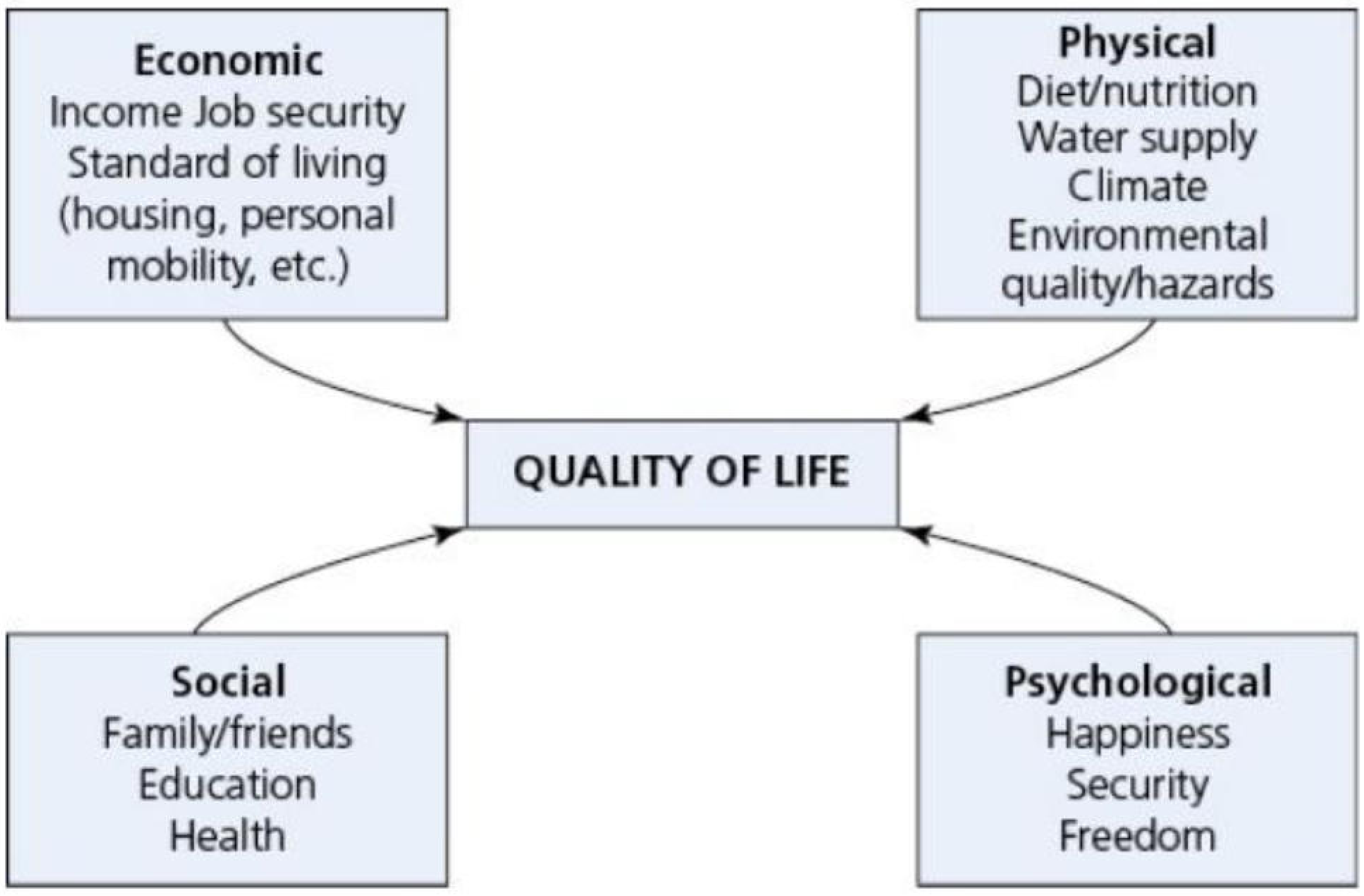
8.1 Measuring development

This chapter will explain:

★ indicators of development
★ the Human Development Index: a broader measure of development.

Indicators of development

Development, or improvement in the quality of life, is a wide-ranging concept. It includes increasing wealth, but also involves other important aspects of our lives (Figure 8.1). Many people would consider a good health service to be more important than wealth. People who live in countries that are not democracies, where freedom of speech cannot be taken for granted, often envy those who do live in democratic countries, though many African countries look to China as a model of success.



▲ **Figure 8.1** Factors comprising the quality of life

Development occurs when there are improvements to the individual factors making up the quality of life and standard of living (Figure 8.2). The more factors that are improved in terms of people's wellbeing, the greater the rate of overall progress. For example, development occurs in a low-income country when:

» local food supply improves due to investment in farm machinery and fertilisers
» the electricity grid extends outwards from the main urban areas to rural areas
» levels of literacy improve throughout the country
» a new road or railway improves the accessibility of a remote province.

▲ **Figure 8.2** An open-pit toilet in the Gobi Desert: in some parts of the world sanitation can be very basic

Thus, development is a process of change for the better. However, in some cases it can proceed in a way that adversely affects the most impoverished in a society. Development is not always a continuous process as it can be interrupted by major national and global events such as war, famine, disease and economic recession. The Covid-19 pandemic had a massive negative impact on development, and set back progress in LICs by many years.

The **United Nations Development Programme (UNDP)** is the UN's development agency working to eradicate poverty. It plays a crucial role in helping countries achieve the Sustainable Development Goals. The UNDP has highlighted a number of important issues concerning development:

» Development is about creating opportunities and giving people the freedom to make choices.
» Sustainability is vital for long-term development.
» Development must balance social, economic and environmental sustainability.

Table 8.1 looks at four categories of development, giving examples of how countries can improve these aspects of the quality of life of their citizens. It is not an exclusive list and you may well be able to think of other examples within each category.

▼ **Table 8.1** Categories of development and examples

Category	Examples
Social	● Achieving higher levels of literacy ● Reducing infant and maternal mortality ● Improving life expectancy ● Reducing the gender gap ● Improving the quality of housing and other aspects of soft infrastructure
Economic	● Raising productivity ● Increasing incomes for all levels of society ● Improving hard infrastructure (roads, railways, airports) ● Achieving a better balance between economic sectors ● Increasing trade with other countries ● Raising per capita energy consumption ● Increasing level of car ownership
Environmental	● Reducing CO_2 and other greenhouse gas emissions: cutting lower atmosphere air pollutants ● Improving water quality in rivers and other water bodies ● Tackling land pollution and other issues such as deforestation ● Conserving biodiversity
Political	● Improving democratic institutions ● Implementing measures to reduce corruption ● Keeping to new international human rights obligations

Economic indicators of development

There are three widely used economic indicators of development:

» Gross domestic product (GDP) is the total value of goods and services produced within a country's borders, regardless of the nationality of those involved.
» Gross national product (GNP) focuses on the income generated by a country's residents, regardless of location.
» Gross national income (GNI) provides the most comprehensive picture by incorporating both GDP and GNP.

The World Bank, the Human Development Index and the annual World Population Data Sheet published by the Population Reference Bureau all use GNI to illustrate the development differences between countries. For all three of these economic indicators, both absolute and relative data is available. Relative data, for example **GNI per capita**, tends to be the most widely used. GNI per capita (see Table 8.2) is calculated by dividing the total GNI of a country for a given year by the number of people in that country. Per capita data allows comparisons to be made between countries that have big differences in their population size. For example, the total GNI of China is greater than that of France, but GNI per capita is much higher in France.

However, such data do not take into account the differences in the cost of living between countries. For example, a dollar buys much more in Bangladesh than it does in the USA. To account for this, the GNI per capita at **purchasing power parity (PPP)** can be calculated.

While GNI per capita is a useful measure, it does not tell us anything about:

- wealth distribution within a country – in some countries the gap between highest and lowest incomes is much greater than in others
- the ways in which governments invest the money at their disposal – a number of factors can affect the efficiency with which governments operate, with a high level of **corruption** being a significant problem in some countries.

Economic growth (GNI growth) refers to an increase in real GNI (percentage in a year). 'Real' means after inflation is taken into account. Countries that experience high rates of inflation often struggle to cope with rising prices as average incomes lag behind. A high rate of inflation can adversely affect economic confidence within a country itself and also impact investment coming into the country from abroad.

Activities

1 What is development?
2 Give two examples of development in a low-income country.
3 The UNDP stresses the importance of a balance between which three types of development?
4 What is the reason for calculating GNI data at purchasing power parity?

▼ **Table 8.2** Key development indicators by world region

Region	Life expectancy at birth (years)	Infant mortality rate (per 1000)	GNI per capita (US$)	Per capita kilocalorie supply from all foods per day (kcal)
World	72	29	25,100	2959
Africa	63	44	5645	2573
North America	77	5	75,470	3587
Latin America/Caribbean	74	13	18,412	3108
Asia	74	24	16,349	2927
Europe	78	4	47,960	3456
Oceania	79	15	43,904	3087

Other indicators of development

Many individual indicators other than those above, which are clearly economic, have been used to highlight contrasts in development. Some of the most widely used are the literacy rate, life expectancy at birth, infant mortality, calorie intake and number of people per doctor. Development varies not only between countries – it can also vary significantly within countries. For instance, the number of people per doctor is likely to be higher in rural areas. Most of the measures that can be used to study the contrasts between countries can also be used to look at regional variations within countries.

Literacy

Education is the key to socioeconomic development. It can be defined as the process of acquiring knowledge, understanding and skills. Education has always been regarded as a very important indicator of development and it has figured prominently in aggregate/composite measures of development. Adult literacy (the ability to read and write) is one of the main ways in which differences in educational standards between countries can be shown. In 2020, the global **adult literacy rate** was estimated at 86 per cent. For developed nations it is at least 96 per cent, but for LICs the average adult literacy rate is only 65 per cent. A low adult literacy rate is a great obstacle to development. Poverty and illiteracy go hand in hand. Many of the countries with the lowest literacy rates are in Sub-Saharan Africa.

Of the roughly 780 million adults worldwide who cannot read and write, nearly two-thirds are female. Improving female literacy is essential because so many aspects of development depend on it. For example, there is a very strong relationship between levels of female literacy and infant mortality rates. People who are literate are much more able to access medical and other information that will help them achieve a higher quality of life compared with those who are illiterate.

Life expectancy at birth

Life expectancy at birth is to a large extent the end result of all the factors contributing to the quality of life in a country. The main influences on life expectancy are:

- the incidence of disease, e.g. malaria (Figure 8.3)
- physical environmental conditions, e.g. very low rainfall
- human environmental conditions, e.g. pollution
- personal lifestyle, e.g. smoking.

Rates of life expectancy in HICs and LICs have converged significantly during the last 50 years, in spite of a widening wealth gap. Table 8.2 shows that the lowest life expectancy by world region is in Africa, at 63 years. However, in 2014, Africa's life expectancy was 59 years. It is encouraging that some aspects of development can improve quite quickly!

Infant mortality

The **infant mortality rate** is regarded as one of the most sensitive indicators of socioeconomic progress. It is an important measure of health inequality between and within countries. There are huge differences in infant mortality around the world, despite the wide availability of public health knowledge. The main causal factors of such disparities are:

- contrasts in material resources
- differences in the efficiency of social institutions and health systems.

▲ **Figure 8.3** A public health poster in a park in Buenos Aires, Argentina – disease control is an important aspect of development

Fortunately, infant mortality rates have fallen sharply in many MICs and LICs over the last 30 years. Nevertheless, Table 8.2 shows that the infant mortality rate in Africa is 11 times that in Europe. Sixteen countries had an infant mortality rate of 50/1000 or more in 2023 – this means one in 20 infants dying before their first birthday. All 16 of those countries are in Sub-Saharan Africa, with the highest levels of infant mortality being in Sierra Leone (75/1000) and Lesotho (69/1000). Table 8.3 shows the extent to which infant mortality varies by region in Africa. Some countries have registered very large reductions in infant mortality in recent decades, while others have seen reductions at a lower rate.

▼ **Table 8.3** Infant mortality rate by region in Africa, 2023

Region	Number of countries above 50/1000	Average infant mortality rate
Northern Africa	0	23
Western Africa	8	54
Eastern Africa	3	39
Middle Africa	4	51
Southern Africa	1	27

Infant mortality generally compares well with many other indicators of development, which is a good indication of its value as a measure of development.

Daily calorie intake

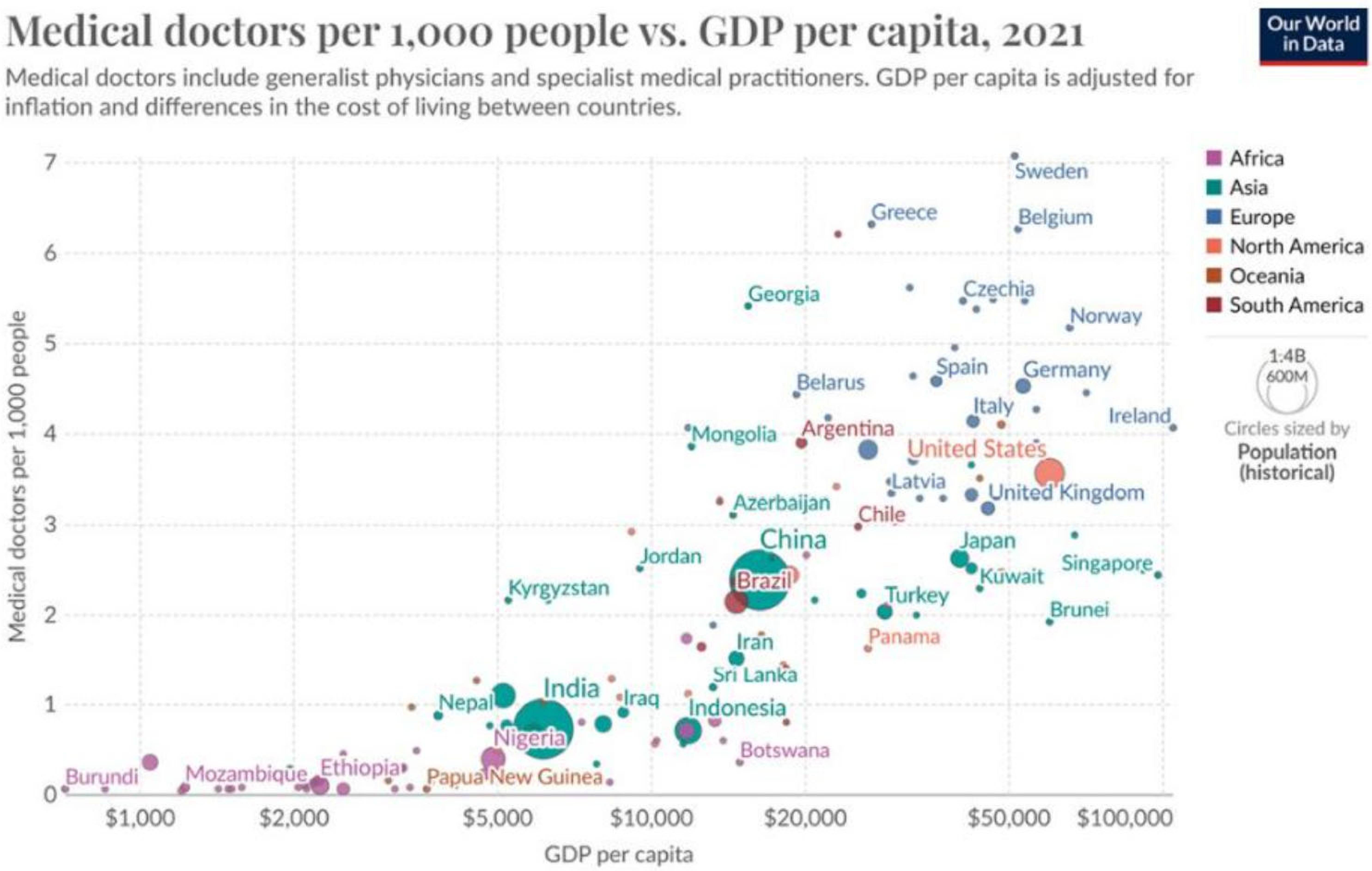
Daily calorie intake is viewed as an important measure because people need energy, which is provided by calories, for their bodies to function properly. While the average figures by world region are useful in showing general contrasts around the world, it is important to remember that the countries with the lowest average calorie intake are those that usually suffer the greatest degree of food insecurity. During times of drought and famine, daily calorie intake may only be a fraction of the 'average' figure. A general daily guideline for weight maintenance is 2000 calories for women and 2500 for men. However, excessive calorie intake can result in obesity and associated metabolic diseases.

Doctors per thousand people

The number of doctors per unit of population is an indication of the financial resources available to a country for the training and recruitment of doctors. Access to well-qualified medical professionals impacts directly on the wellbeing of patients. Figure 8.4 is a scatter graph that shows the relationship between medical doctors per thousand people and GDP per capita for a wide range of countries. The graph shows that most, but not all, HICs have more than three doctors per thousand people, while LICs generally have less than one. In fact many LICs are in the situation that they are nowhere near one doctor per thousand people – look how many LICs are very close to the baseline (the horizontal axis) on the diagram. Overall, the scatter graph shows that there is a strong correlation (relationship) between doctors per thousand people and GDP per capita.

Interesting note

Corruption is a major obstacle to economic and social development around the world. The Corruption Perceptions Index has been published annually since 1995 by Transparency International. It ranks the countries of the world for perceived levels of public sector corruption.



▲ **Figure 8.4** The relationship between doctors per 1000 people and GDP per capita

The relationship between different indicators of development

All indicators of human development have merits and limitations. The indicators discussed above have been widely used because their merits are generally considered to be greater than their limitations. Therefore, it is reasonable to assume that you might expect there to be a strong (but not perfect) relationship between any pair of these indicators, such as that illustrated by Figure 8.4.

Activities

1 Define adult literacy.

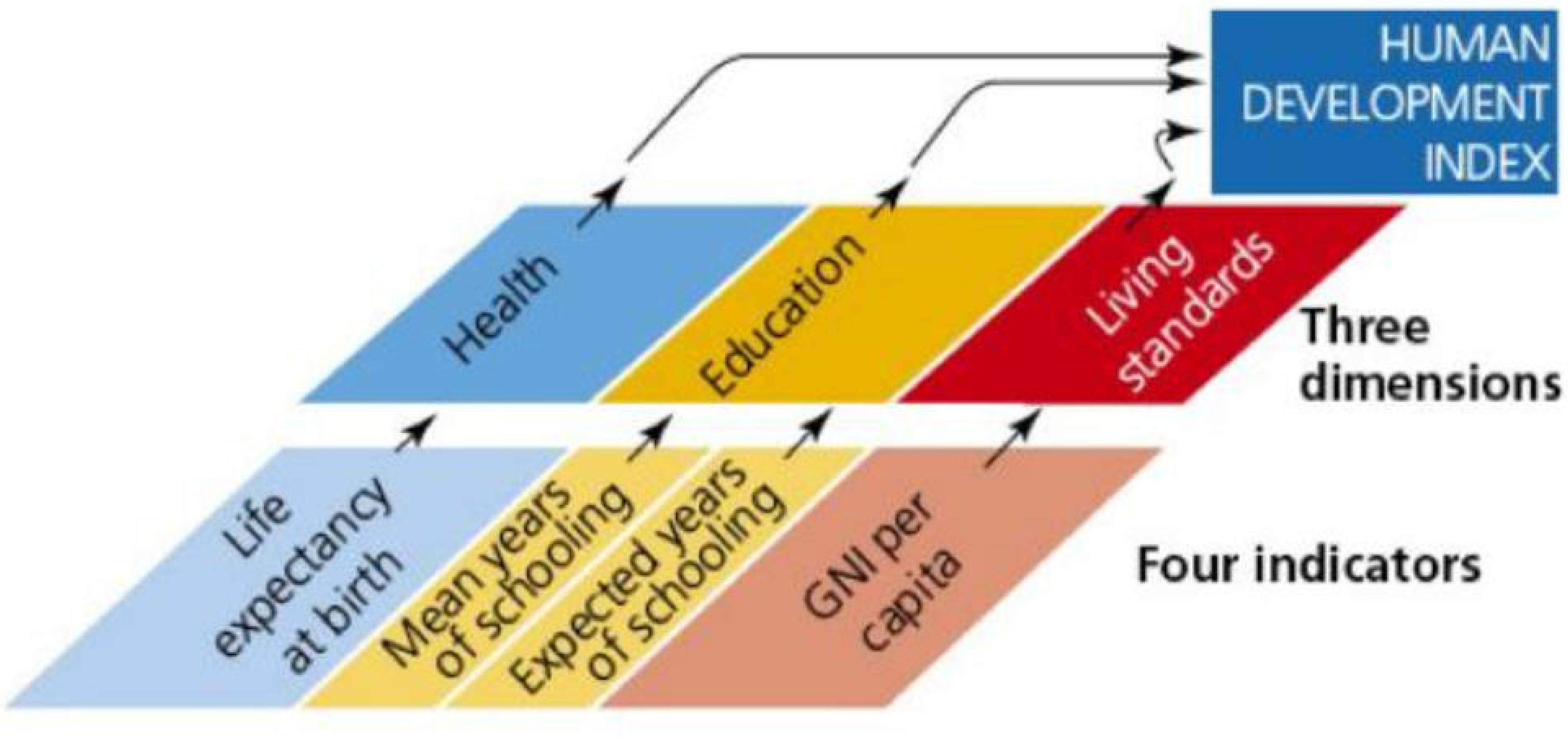
2 Which world regions have the highest and lowest infant mortality rates?
3 List the four main influences on life expectancy.
4 Using the data from Table 8.2 (page 180), draw a scatter graph to show the relationship between life expectancy at birth and infant mortality rate. Insert a line of best fit and comment on the relationship between the two variables.

The Human Development Index: a broader measure of development

The **Human Development Index (HDI)** was devised by the United Nations in 1990. It is a composite index, combining a number of key development indicators. Its aim was to extend the concept of development beyond wealth alone. The United Nations Development Programme (UNDP), which is responsible for producing the HDI, has stated that 'The HDI is a summary measure of average achievement in key dimensions of human development – a long and healthy life, being knowledgeable and having a decent standard of living'. Canada (Figure 8.5) has a very high level of human development by these measures, but the UNDP acknowledges that the HDI includes only part of what human development entails. For example, it does not include measures of inequality, poverty, human security or empowerment.

▲ **Figure 8.5** The Waterfront, Vancouver: Canada has a very high level of human development

The HDI has changed slightly in composition over the years. The current index combines four indicators of development (Figure 8.6):

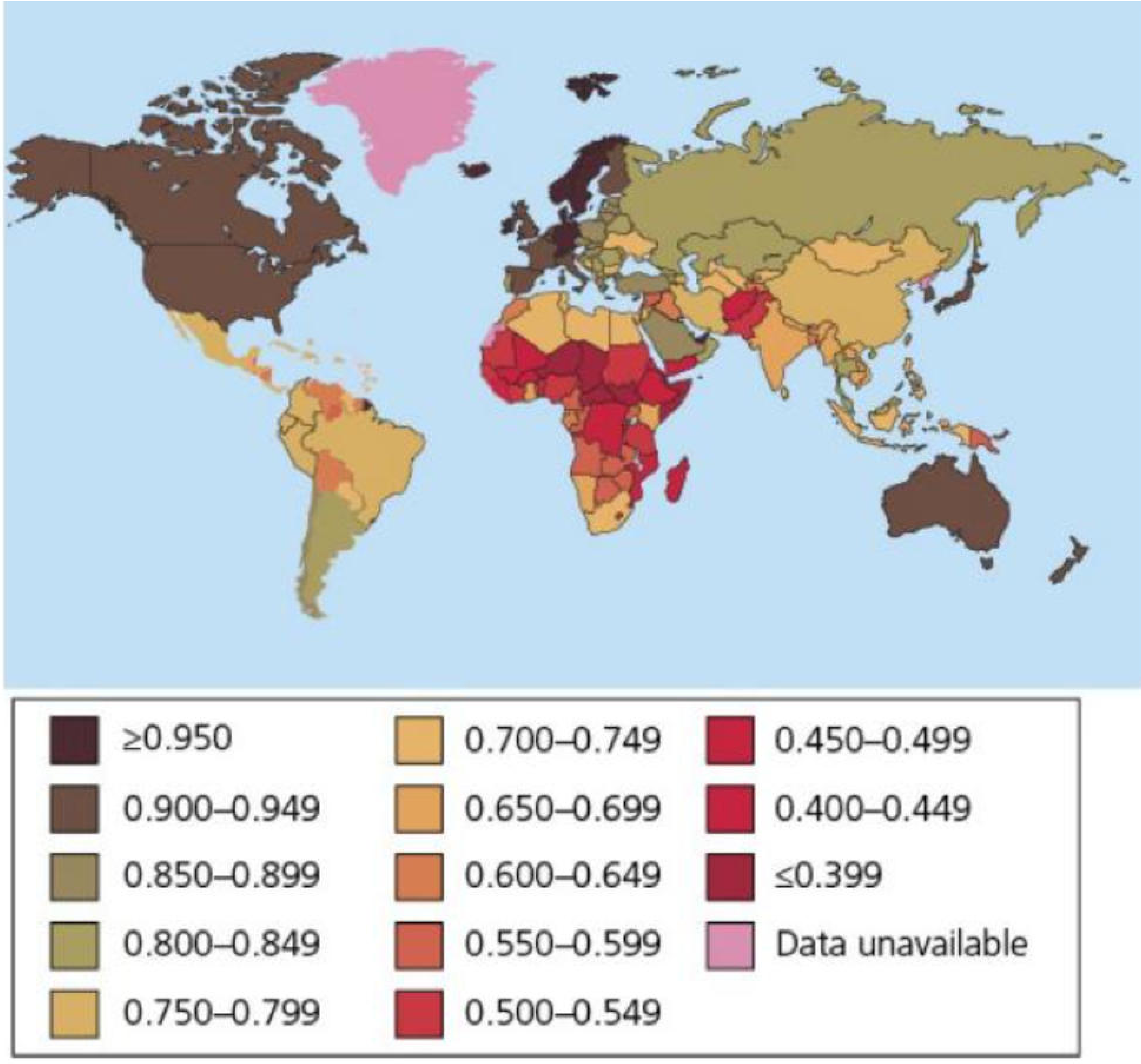


▲ **Figure 8.6** The components of the Human Development Index

- Life expectancy at birth
- Mean years of schooling for adults aged 25 years
- Expected years of schooling for children of school-entering age
- GNI per capita ($PPP).

The actual figures for each of these measures are converted into an index, which has a maximum value of 1.0 in each case. The index figures are then combined and averaged to give an overall HDI value. This also has a maximum value of 1.0 (Figure 8.7). The United Nations publishes an updated Human Development Report every year, which uses the HDI to rank all the countries of the world on their level of development. The HDI divides the countries of the world into four groups. The 2024 report, based on 2022 data, divided the countries of the world as follows:

- Very high HDI: The countries ranked 1–69
- High HDI: The countries ranked 70–115
- Medium HDI: The countries ranked 116–159
- Low HDI: The countries ranked 160–193.



▲ **Figure 8.7** The Human Development Index scores in increments of 0.050, from the 2024 report, based on 2022 data

Table 8.4 shows examples of countries in each group in the 2024 report. Some countries have progressed from one group to another in a relatively short period of time.

▼ **Table 8.4** Examples of countries in each group in the 2024 report

Level of development	Example countries
Very high human development	Norway, Germany, Canada, UK, Japan, Australia
High human development	Cuba, Mexico, Brazil, China, South Africa, Indonesia
Medium human development	India, Morocco, Ghana, Pakistan, Philippines, Bangladesh
Low human development	Nigeria, Tanzania, Afghanistan, DR Congo, Yemen, Mozambique

Activities

1. Which indicators of development are combined to form the Human Development Index?
2. Look at Figure 8.7 and briefly describe the distribution of countries according to the Human Development Index.

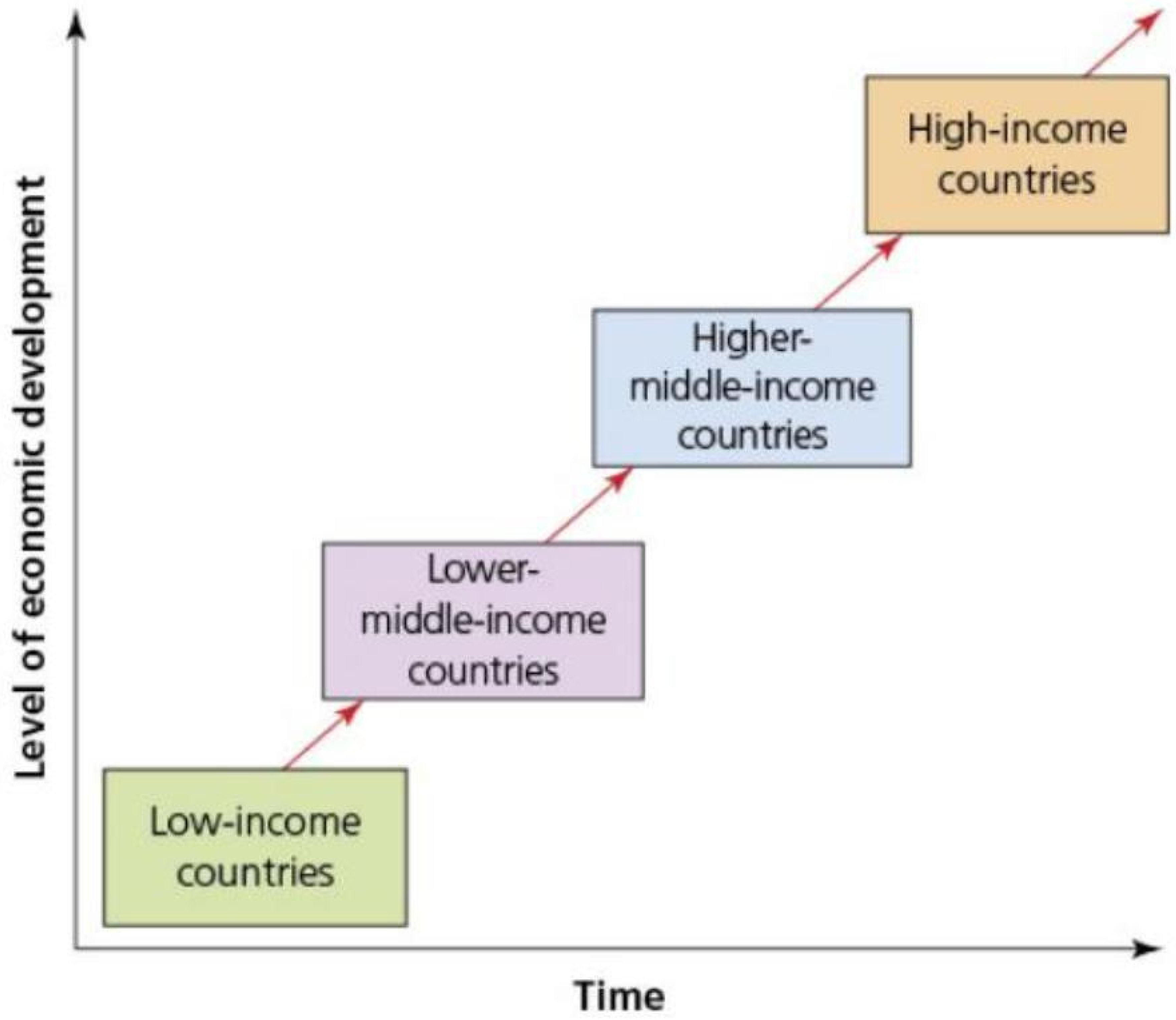
8.2 The world is developing unevenly

This chapter will explain:

★ the global pattern of LICs, MICs and HICs
★ the development gap.

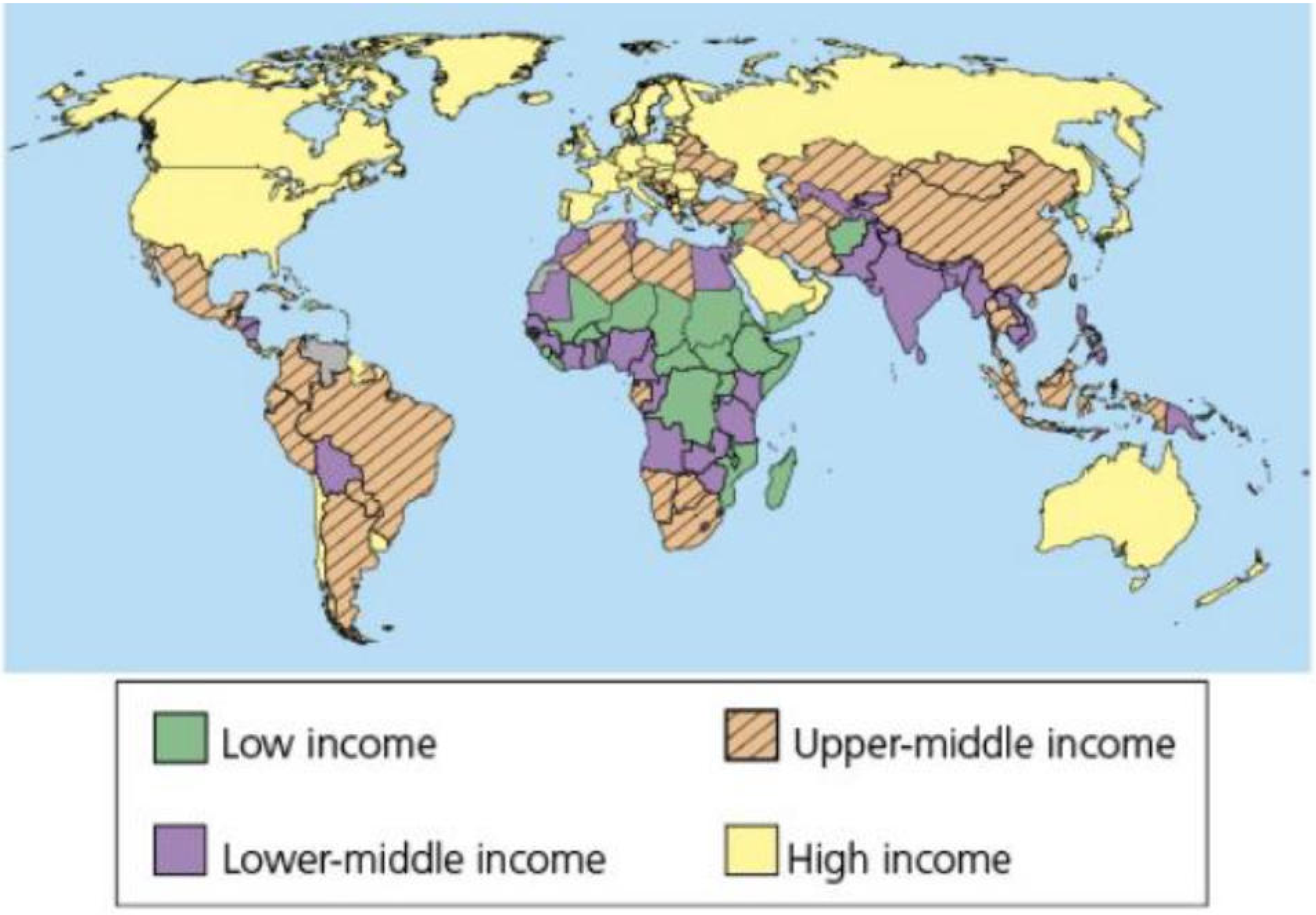
The global pattern of LICs, MICs and HICs

A reasonable division of the world in terms of economic development is shown in Figure 8.8, which identifies three broad global income groups – low-income countries (LICs), middle-income countries (MICs) and high-income countries (HICs). The MICs group of countries is often subdivided (as it is in Figure 8.8) into lower-middle income and upper-middle income. This is because of the very wide range of income and the number of countries involved in the MIC category.



▲ **Figure 8.8** The stages of development

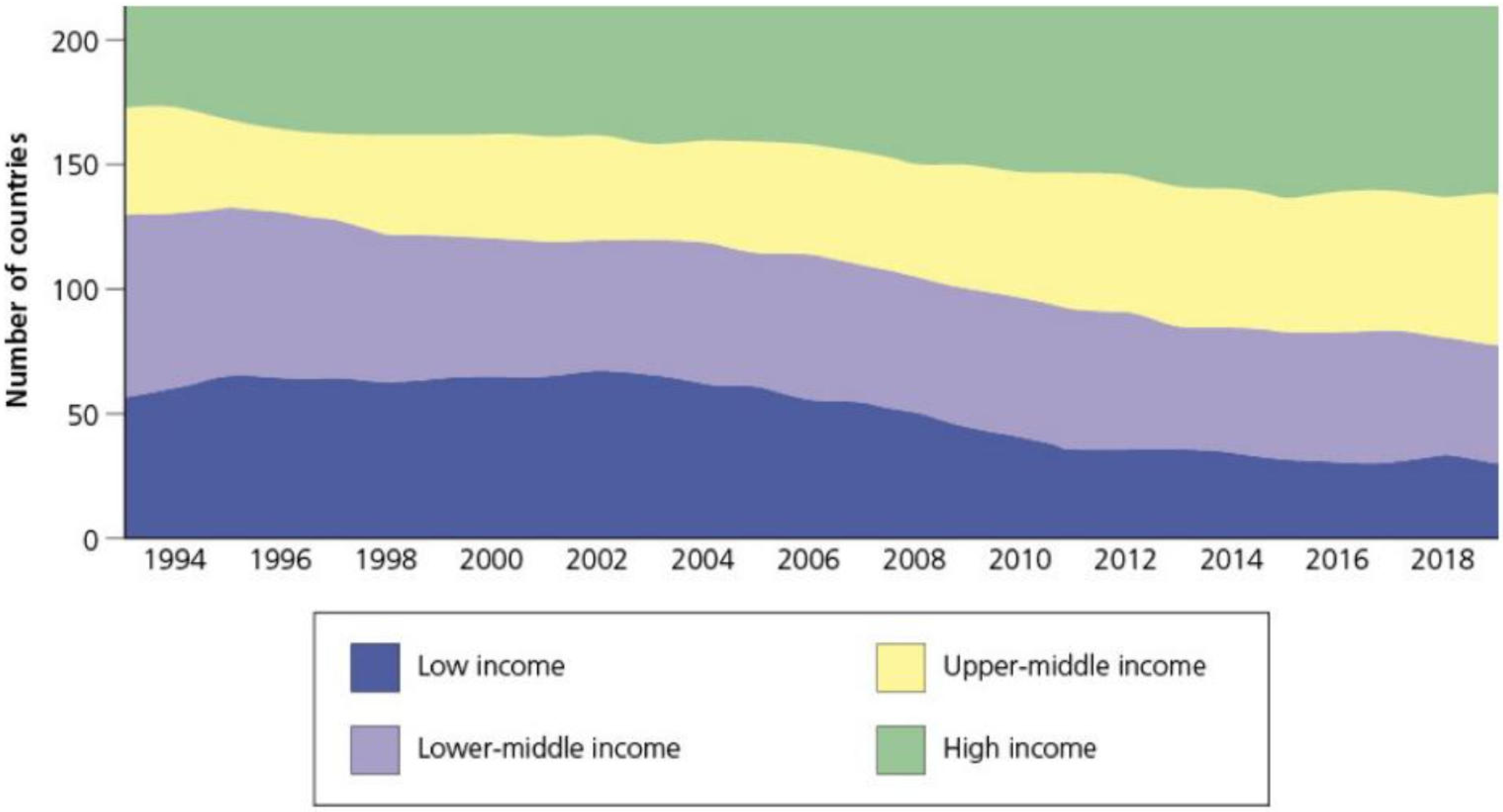
As countries develop economically and socially they can move up the '**ladder of development**', but under significant adverse circumstances they can also move down. Figure 8.9 shows the global pattern of countries by income group published by the World Bank in 2024, using 2022 data. The World Bank uses GNI per capita data in US dollars, converted from local currency using the World Bank Atlas method, which averages out exchange rate fluctuations. Examples of countries in each income group are given in Table 8.5.



▲ **Figure 8.9** The global pattern of countries by income group

▼ **Table 8.5** Examples of countries in each income group in 2024

Income group	GNI per capita	Example countries
High income	$13,846 or more	USA, Canada, Germany, UK, Japan, Australia
Upper-middle income	Between $4466 and $13,845	Brazil, Argentina, Turkey, China, Indonesia, South Africa
Lower-middle income	Between $1136 and $4465	Bolivia, Algeria, Nigeria, India, Iran, Vietnam
Low income	$1135 or less	Ethiopia, Afghanistan, Sudan, DR Congo, Mozambique, Somalia



▲ **Figure 8.10** Changes in the number of countries by income group, 1993–2019

Figure 8.10 shows that between 1993 and 2019:

» The number of high-income countries increased considerably.
» The number of low-income countries declined sharply, particularly from the early 2000s.
» The number of middle-income countries changed to a much lesser extent, as countries transitioned both into and out of this group.

In the 1990s, more than six in 10 people in the world lived in low-income countries. Today it is less than one in 10. However, critics of the World Bank system of income classification say that with only a small proportion of the world's population classified as low income and as income thresholds have not changed since 1988 (in real terms), these income groups are somewhat arbitrary and dated.

Low-income countries

LICs are located predominantly in Africa, with most of the rest in Asia (Figure 8.9). With a total population of over 600 million, LICs accounted for just over 1.1 per cent of global GNI in 2023. In the same year, LICs contributed about 1 per cent to world trade. Table 8.6 shows World Bank data comparing LICs with MICs and HICs on a range of indicators.

▼ **Table 8.6** World Bank data comparing LICs, MICs and HICs

Indicator	Year	LICs	MICs	HICs
Population total (billion)	2023	0.723	5.9	1.4
Population growth (% p.a.)	2023	2.7	0.8	0.5
Poverty ratio (% pop.)	2022	44.5	N/A	0.3
Life expectancy at birth (years)	2022	63	71	80
Unemployment (%)	2023	5.3	5.1	4.4
CO_2 emissions (tonnes per capita)	2020	0.3	3.7	7.8
Access to electricity (% pop.)	2022	44.9	94.9	100.0
People using safely managed sanitation services (% pop.)	2022	24	55	91

LICs are often **primary product dependent** (commodity dependent). This occurs when a country relies on one or a small number of primary products for most of its export earnings. As the gap between the richest and poorest countries in the world widens, LICs are being increasingly **marginalised** in the world economy. Marginalisation is the process of being pushed to the edge of global economic activity and of being largely left out of positive economic trends. The term can be applied to countries, to regions within countries, and to groups within a population – the poorest, the unemployed, the most vulnerable (Figure 8.11). There are great concerns that marginalisation has increased under the process of globalisation. The problems of LICs are often made worse by major geographical handicaps and natural and human-made disasters. Examples include the following:

» The catastrophic earthquake in Haiti in 2010 that killed an estimated 220,000 people and caused massive damage. Another severe earthquake hit Haiti in August 2021.
» Bangladesh experiences flooding every year, and when flooding is severe it can impact 75 per cent of the country. About three-quarters of Bangladesh is less than 10 metres above sea level.
» Small Island Developing States (SIDS) are the most disaster-prone countries in the world. They are regularly impacted by severe storms and other natural hazards. According to the United Nations Conference on Trade and Development (UNCTAD), this causes on average an annual damage of 2.1 per cent of GDP.

▲ **Figure 8.11** Poor transport infrastructure in remote regions is a major obstacle to development

The UN has called for increased investment in the **productive capabilities** of the LIC countries. This refers to the productive resources, entrepreneurial capabilities and production links that together determine a country's capability to produce goods and services, and to enable it to grow and develop.

LICs have experienced more frequent instances of **growth collapse** than other groups of countries. Growth collapse is a significant decline in economic activity brought about by factors such as a global economic shock or by the impact of a major natural hazard. Covid-19 is the latest example of a growth collapse, reversing progress achieved on several development dimensions.

Middle-income countries

Middle-income countries (emerging economies) are nations that have moved up the development ladder, having at different times in the past been considered LICs. MICs account for about 75 per cent of the world's total population and around 62 per cent of those in poverty. Representing about a third of global GDP, the most dynamic MICs are the fastest-growing economies in the world.

The first countries/territories to make the transition from low to middle income (in the 1960s) were South Korea, Singapore, Taiwan and Hong Kong (Figure 8.12). The media referred to them as the 'four Asian tigers'. A '**tiger economy**' is one that is growing rapidly. The reasons for the success of these countries/territories were:

- a good initial level of infrastructure
- a skilled but relatively low-cost workforce
- cultural traditions that revere education and achievement
- governments that welcomed foreign direct investment from transnational corporations
- distinct advantages in terms of geographical location
- the ready availability of bank loans for businesses, often extended on government orders and at attractive interest rates.

▲ **Figure 8.12** Hong Kong was one of the newly emerging economies that began to develop in the 1960s

The success of the four Asian tigers provided a model for others to follow, such as Malaysia, Brazil, China and India. South Korea and Singapore have developed so much that they are now considered to be HICs.

High-income countries

The high-income group of countries covers an interesting range. The core of this group is the traditional developed countries, long recognised by the United Nations – Northern America, Western Europe, Japan, Australia and New Zealand. It also includes the following countries/territories:

- Emerging economies/newly industrialised countries that have developed since the 1960s, so far and so fast that they have reached high-income status. Examples are Singapore, South Korea and Hong Kong.
- Small island nations within the European Union. Examples are Cyprus and Malta.
- Eastern European countries that have joined the European Union since the disintegration of the Soviet Union.
- Countries that have become rich through the exploitation of natural resources, particularly oil and gas. Examples are the UAE, Bahrain and Guyana.
- Tax havens such as the Cayman Islands and the British Virgin Islands.
- Small countries/territories where tourism has expanded rapidly and profitably, such as the Seychelles and Aruba.

Activities

1 **a** Define primary product dependency.
  b What are the disadvantages of a country being primary product dependent?
2 What do you understand by the terms:
  a productive capabilities
  b growth collapse?
3 Compare LICs, MICs and HICs for three of the indicators in Table 8.6 (page 185).
4 **a** What is a tiger economy?
  b Identify the factors responsible for the development of the first generation of tiger economies in the 1960s.
5 In what ways do some of the HICs identified by the World Bank differ from the traditional developed countries, long recognised by the United Nations?

Explaining the development gap

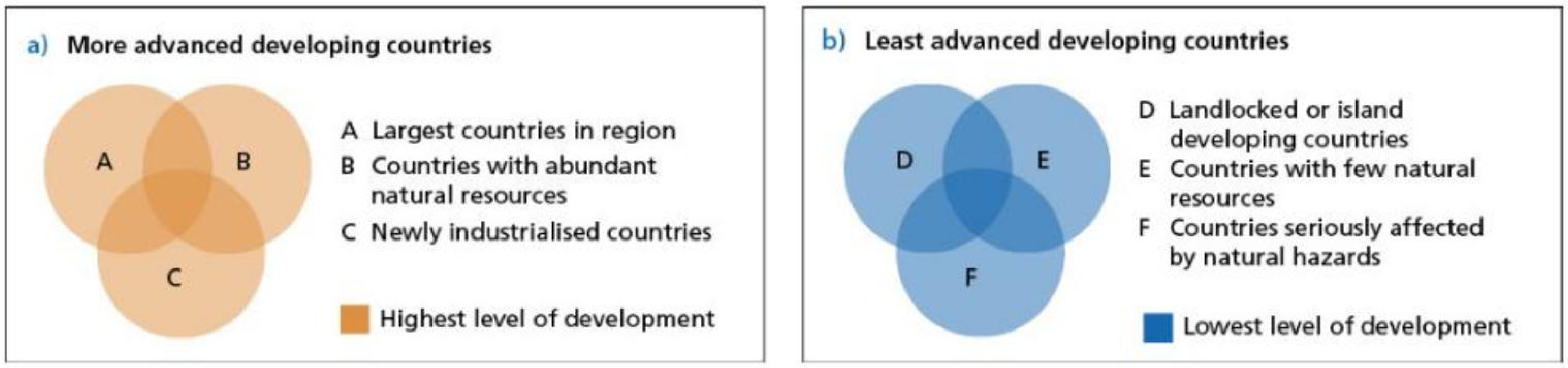
There has been much debate about the causes of development and the extent of the **development gap**.

The variations between countries are due to a range of factors. Physical geography can have a major impact on development.

» **Landlocked countries** (for example Niger and Nepal) have generally developed more slowly than coastal nations. Lacking direct access to sea ports increases the cost of importing and exporting goods as these need to be transported through one or more neighbouring countries.
» Small island countries (for example Kiribati and São Tomé and Principe) face considerable disadvantages in development. They often have limited raw materials and are highly dependent on tourism. Most are also highly vulnerable to the effects of climate change.
» Tropical countries have grown more slowly than those in temperate latitudes, reflecting the costs of poor health and unproductive farming in more challenging physical environments. Poor health is a result of high prevalence rates of a range of tropical diseases, along with poor nutrition due to low agricultural productivity.
» A generous allocation of natural resources has spurred economic growth in a number of countries, but there are examples where resource wealth has been wasted (for example Nigeria, a major oil producer).

Appropriate, realistic and well-applied economic policies can be an important factor in the development process:

» **Open economies** that encouraged foreign investment have developed faster than closed economies. Some MICs and LICs have established special economic zones to attract foreign direct investment. However, political manipulation by developed nations has sometimes caused developing nations to open their economies too fast or in other unhelpful/unsustainable ways.
» Fast-growing countries tend to have high rates of saving and low spending relative to GDP. A high rate of saving can provide money for investment in infrastructure for transport, energy, health, education, housing and water supply.



▲ **Figure 8.13** Venn diagrams showing fast and slow development in developing countries

» **Institutional quality** in terms of good government, law and order, and lack of corruption generally result in a high rate of growth. These positive attributes also encourage foreign direct investment.

Progress through demographic transition is a significant factor, with the highest rates of economic growth experienced by those nations where the birth rate has fallen the most, due to women being able to access education and be included in the workforce.

Figure 8.13 combines a range of factors to explain differences in development. For example, in diagram (a), Brazil would satisfy all three criteria. It is by far the largest country in South America, it has abundant natural resources, and it is an emerging economy. In contrast, countries such as Haiti and Niger would be affected by all three of the negative factors in diagram (b), based on a history of colonisation, which affected their land and borders, infrastructure development and debts.

Colonialism and neo-colonialism

Colonialism occurs when one country takes control of another and dominates it politically and economically. It involves people from the colonial power invading and settling in the newly controlled country. The age of 'modern colonialism' began around 1500, following the European discovery of both America and a sea route around Southern Africa. Portugal and Spain dominated the early period of colonialism, but were later followed by the Netherlands, Britain, France, Germany and Belgium.

Colonialism spread at a particularly fast pace in the eighteenth and nineteenth centuries. A period of special note was the 'Scramble for Africa' (or the Partition of Africa) between 1880 and the beginning of the First World War (1914). This involved the colonisation and division of most of Africa by the seven European powers named above. By the early twentieth century, a large majority of the rest of the world had been colonised by the European powers at some point in time. These actions by colonial powers brought about massive negative changes in many countries. Resistance to the colonisers was overwhelmingly met with violence and intimidation.

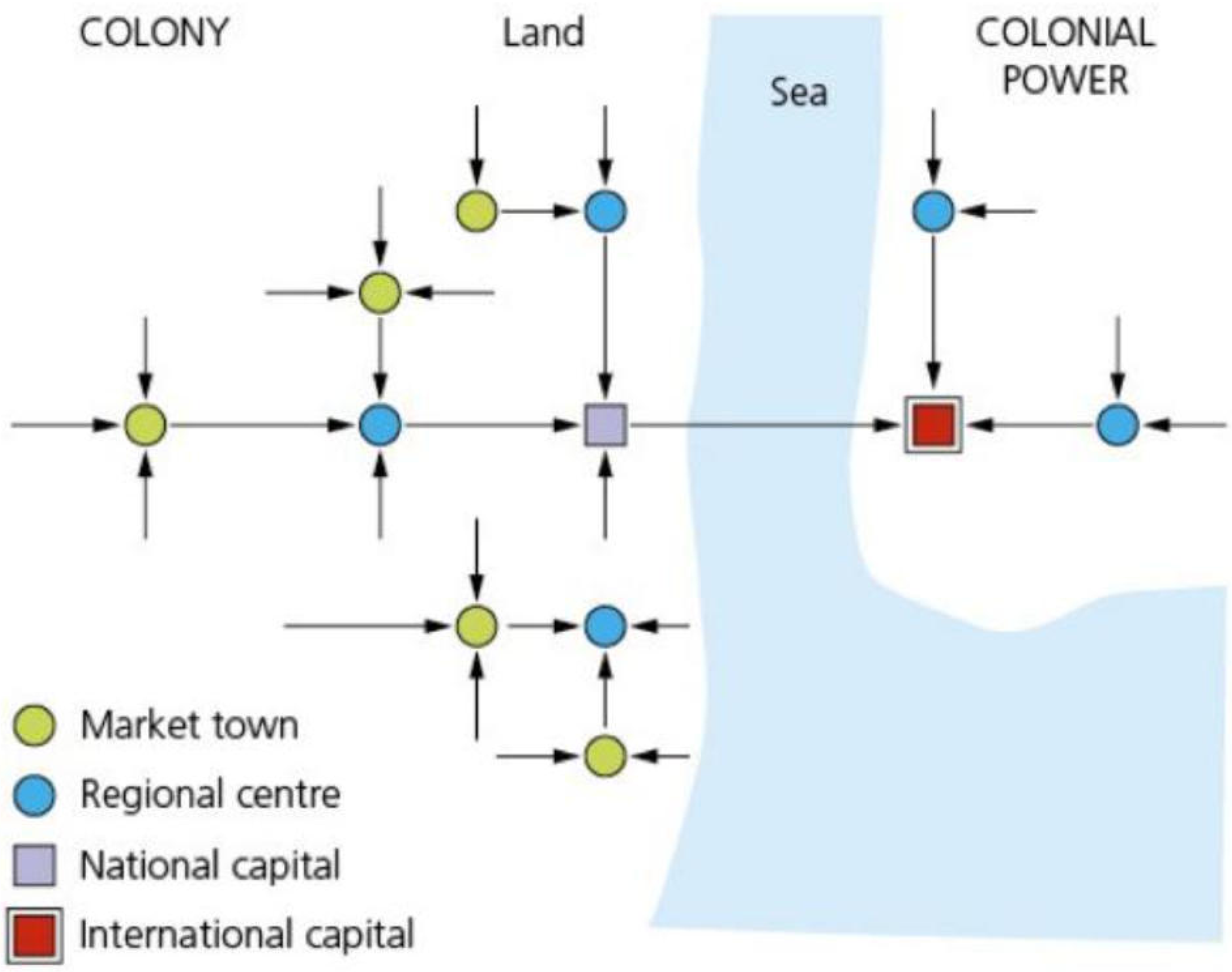
The European countries exploited their colonies for economic gain (both in terms of raw materials and labour), causing enormous suffering and future damage as a result of their actions. The imposition of unequal trading relationships has had a long-lasting effect. However, slavery was undoubtedly the most brutal aspect of colonialism.

In the modern world, the term '**neo-colonialism**' is used to describe how HICs can still dominate LICs in an economic and political sense.

Dependency theory was developed to help explain the impact of these processes. In the late 1960s, the economist A.G. Frank produced a theory that was critical of **capitalism**. He argued that:

- Poverty in the developing world was not an original condition, but that it developed through the spread of world capitalism. Many countries had been prosperous before the arrival of European colonists.
- The process of absorption into the capitalist system sowed the seeds of underdevelopment.
- The development of European and other dominant countries was achieved by the extraction of resources from developing countries.
- Developing countries became even more dependent on the West through substituting the production of export goods where once local food crops prevailed.
- The stronger the links to the developed world, the worse the level of development in LICs.

Frank used a simple model to explain how the '**economic core**' (the developed world) exploited the '**economic periphery**' (the developing world). His model (Figure 8.14) shows a chain of exploitation. It begins with small towns in the periphery exploiting resources from their surrounding rural areas. This process of exploitation works its way up the urban hierarchy in the periphery until, finally, the largest settlements are exploited by cities in the colonial power. The intensity of poverty increases with the number of stages down the chain of exploitation. As with most theories that try to explain complex situations, dependency theory has its advocates and critics.



▲ **Figure 8.14** Dependency theory

The roles of trade and investment in development

Trade and investment play a key role in economic development. Investment in a country is the key to it increasing its trade. Some developing countries have increased their trade substantially in recent decades. Examples are China, India, Brazil and Mexico. Such countries have attracted the bulk of foreign direct investment. The emergence of newly industrialised countries (emerging economies) has been the biggest success of globalisation.

However, about 2 billion people live in countries where trade has fallen in relation to national income. In a sense, these countries have become less globalised rather than more! The share of world trade for many of the world's low-income countries has fallen in recent decades. Non-governmental organisations (NGOs) argue strongly that trade is the key to real development, being worth many times more than international aid. These NGOs continue to criticise the ways in which world trade operates to the detriment of developing countries. Sometimes, when a developing country increases the volume of its exports, its revenues have declined due to a fall in prices on the world market.

Political and economic policies

Open economies such as the UK and Singapore that encourage foreign investment have developed faster than countries that have much greater controls on foreign investment. Large-scale investment can create significant economic **multiplier effects**. Closed economies engage in only limited trade and other contacts with the outside world. Their development tends to suffer as a consequence. North Korea is the most extreme example of a closed economy. However, the strong market **protectionism** practised by some Western countries undermines them being perceived as open economies.

Until the early 1990s, India was a relatively closed economy. There were very high tariffs on imports, along with other restrictions. Reducing barriers to trade was an important part of the economic reforms the country made at the time. As India became more integrated into the global economy, the volume of both its exports and its imports rose sharply.

Social investment

Countries that have prioritised investment in education and health have generally developed at a faster rate than nations that have invested less in these sectors. High standards of education and health are vital for sustainable development. When Sierra Leone gained independence in 1963, it reduced investment in its health system. Since then the country's economy has struggled.

In contrast, Botswana has been described by the UNDP as 'one of Africa's veritable economic and human development success stories'. Instead of squandering the wealth generated by its diamond and tourist industries, it invested heavily into health and education. Primary school attendance up to age 13 is almost 90 per cent and 85 per cent of the population now live within 3 miles of a health facility. A healthy and well-educated population attracts investment and therefore encourages development. However, even affluent countries can struggle to provide enough investment for healthcare, as the Covid-19 pandemic illustrated.

As countries develop, fertility declines. Greater availability and knowledge of contraception and family planning is key to lower fertility. In addition, female education can create greater social awareness and opportunities for employment.

Technology and development

There are big differences between countries in their capacity to create and use technology for economic growth and development. Technology has significant potential to help deliver the Sustainable Development Goals (see page 192). However, it can also be at the root of exclusion and inequality. The development of human skills is fundamental to technological capacity. It is important to harness the benefits of advanced technologies for all. **Technology transfer** from developed nations to developing countries can play an important role in reducing the technology gap and aiding the development process.

Digital technology has the potential to tackle poverty by providing access to basic services such as e-health, online education and business advice. It can also be used by governments to better connect to their populations through e-government tools. The number of internet users worldwide increased from 2.08 billion in 2012 to 4.95 billion in 2022. The '**digital divide**' is a term used to refer to inequalities in access to ICT. Such inequality can be seen between:

» developed and developing countries
» urban and rural areas
» ethnic and socioeconomic groups in the same country
» different age groups
» males and females.

The **internet penetration rate** is the percentage of a total population that use the internet. It is a measure often used to illustrate the technology gap between world regions (Table 8.7). For individual countries in 2022, South Korea, the UK and Switzerland all reported internet penetration rates of 98 per cent. However, it is not just internet connection in itself, but also the speed of connection that has become increasingly important.

▼ **Table 8.7** The internet penetration rate by world region, 2022

Region	Estimated internet penetration rate, 2022 (%)
World	66.2
Africa	43.1
Asia	64.1
Oceania	70.1
Middle East	76.4
Latin America/Caribbean	80.4
Europe	88.4
North America	93.4

Culture and development

In recent decades, **culture** has come to be viewed as an important factor in development. Culture can be defined as the total of the inherited ideas, beliefs, values and knowledge that constitute the shared basis of social action. It is the way of life of a particular society or group of people. For example, materialism and consumer culture are clearly more important in some countries than in others. Societies that are largely secular may appear to have different priorities to those that are not.

A 2018 British Council report stated, 'When people or communities are given the opportunity to engage with, learn from and promote their own cultural heritage, it can contribute to social and economic development.' A number of UN reports have highlighted the important contribution of indigenous cultures to sustainable development. Respect for cultural diversity helps to promote inclusive societies. The cultural and creative sector of an economy can make an important contribution to GNI and employment. It can increase the attractiveness of a location as a destination to live, visit and invest in.

Activities

1 What are the particular disadvantages for development in landlocked countries and small island countries?

2 **a** Define neo-colonialism.

b How can neo-colonialism hinder development in LICs?

3 Briefly discuss the importance of trade and investment to economic development.

4 Describe social investment.

5 Draw a diagram to illustrate dependency theory.

6 How important is access to modern technology for the development process?

8.3 Achieving sustainable development

This chapter will explain:

- ★ what sustainable development is
- ★ the strategies to try to achieve sustainable development
- ★ the techniques used by selected sectors that promote sustainability
- ★ the strategies to reduce uneven development
- ★ detailed specific example: Indonesia: an emerging economy.

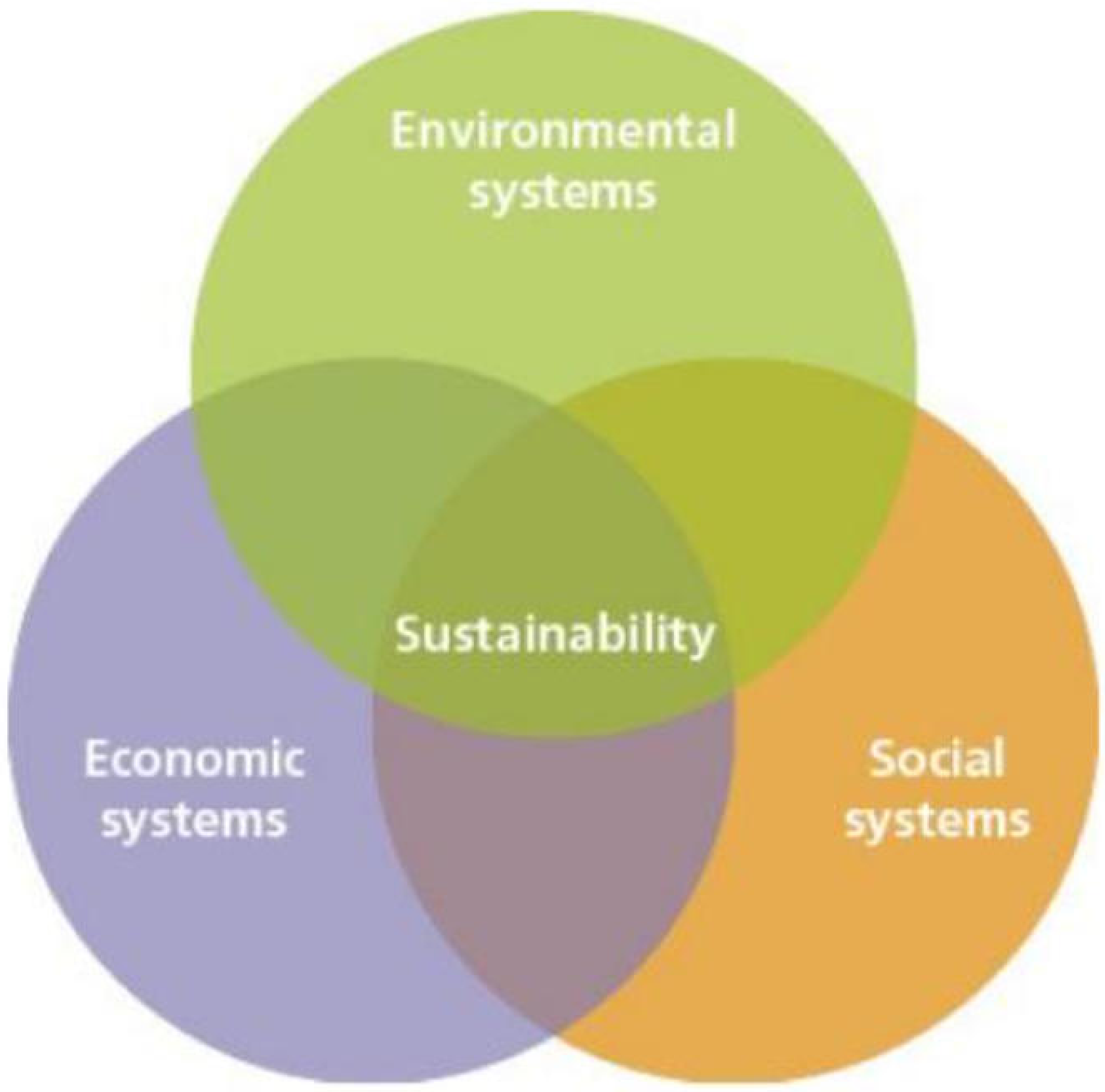
What is sustainable development?

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. The concept of sustainability was established at the first UN Conference on the Environment in 1972, but it did not really take shape until 1987 when the report 'Our Common Future' identified in detail the objectives of sustainable development.

The concept of sustainability can be applied in various ways:

» On the full range of scales, from the individual to the world as a whole. Governments are increasingly reminding individuals and households about their carbon footprint and how this can be reduced. At the largest scale, sustainability focuses on the total carrying capacity of the planet.
» To different geographical environments, such as rainforests, temperate grasslands and urban areas. Satellite photography has enabled a major advance in our ability to see what is happening over large land areas. It has allowed short-term changes to be recognised quickly.
» To individual economic activities such as tourism, agriculture and forestry. Each sector has its own impact on the environment, which can be modified by careful management.

Sustainability need not require a reduction in quality of life, but it does require a change in attitudes and values toward a less consumptive lifestyle. These changes must embrace global interdependence, environmental stewardship, social responsibility and economic viability. Environmental sustainability in a country or region is difficult to achieve without economic and social sustainability because of the strong interconnectedness between these three vital aspects of life (Figure 8.15). Economic sustainability involves maintaining income and employment. Social sustainability means maintaining social capital, including that devoted to health, education, housing and the rule of law.



▲ **Figure 8.15** Social, economic and environmental sustainability

Sustainability is the foundation for the 2030 Agenda for Sustainable Development and its **Sustainable Development Goals**. The UN's 17 Sustainable Development Goals (SDGs) with their 169 targets were adopted by UN Member States in 2015 (Figure 8.16). The overall objectives were to:

a end poverty
b protect the planet
c ensure all people enjoy peace and prosperity.

▲ **Figure 8.16** The 17 Sustainable Development Goals

The 17 SDGs are 'integrated' – recognising that action in one area will affect outcomes in others. The hope was that all these aims would be achieved by 2030. This would go a long way to protect Earth's resilience for future generations.

However, the 2024 Report on Progress towards the SDGs painted a pessimistic picture:

» Only 17 per cent of the 169 SDG targets are on track to be achieved.
» Nearly half are showing only minimal or moderate progress.
» Progress on over a third of the targets has stalled or even regressed.

The 2024 Report on Progress recognised the slow but steady progress in the early years (post-2015) of SDG implementation, but highlighted the mix of global problems since 2019 that have seriously hampered further progress, namely:

» the Covid-19 pandemic
» a growing number of major conflicts
» geopolitical and trade tensions
» the ever-worsening effects of climate change.

All these major issues were additional to the long-lasting problems of:

» significant shortcomings in global economic and financial systems
» historical injustices.

The 2024 Report on Progress views strong global solidarity as vital to achieving meaningful sustainability. It raised serious concerns about the deterioration in global cooperation.

The 2024 Environmental Performance Index

The **Environmental Performance Index (EPI)** uses 58 performance indicators across 11 categories to score and rank countries around the globe on:

» climate change performance
» environmental health
» ecosystem vitality.

These indicators provide a measure at the national scale of how close countries are to reaching environmental policy targets. The 2024 EPI Report concludes that countries' wealth is a strong predictor of overall EPI scores, with good governance being another extremely important factor. In the 2024 EPI overall ranking, European countries occupy the top 20 positions.

▼ **Table 8.8** World map showing rankings in the 2024 Environmental Performance Index

Rank	Country	EPI	Rank	Country	EPI
1	Estonia	75.7	171	Afghanistan	31.0
2	Luxembourg	75.1	172	Iraq	30.3

3	Germany	74.5	173	Madagascar	30.1
4	Finland	73.8	174	Eritrea	29.0
5	United Kingdom	72.6	175	Bangladesh	28.1
6	Sweden	70.3	176	India	27.6
7	Norway	69.9	177	Myanmar	27.1
8	Austria	68.9	178	Laos	26.3
9	Switzerland	67.8	179	Pakistan	25.5
10	Denmark	67.7	180	Vietnam	24.6

Activities

1. Define sustainable development.
2. What does Figure 8.15 (page 191) show?
3. Look at Figure 8.16 (page 192). Research two of the Sustainable Development Goals and then write a brief analysis of your findings.
4. How was the 2024 Environmental Performance Index calculated?
5. Look at Table 8.8. Describe the main characteristics of the countries with the highest Environmental Performance Index (Ranks 1–10) compared to those that have the lowest Environmental Performance Index (Ranks 171–180).

The strategies to try to achieve sustainable development

Environmental

Environmental sustainability centres on the preservation and protection of the natural environment over time. It involves appropriate practices and policies to meet present-day needs without compromising the availability of resources in the future. The key objectives of environmental sustainability are as follows:

- To reduce GHG emissions in all sectors of the economy.
- To transition to renewable sources of energy.
- To adopt sustainable practices in agriculture.
- To increase awareness of the issues involved in environmental sustainability.
- To promote the **circular economy** (Figure 8.17), which aims to keep raw materials in a closed loop. In this way, the maximum use is made of raw materials, resulting in a reduced need for new raw materials. Waste is reduced and the lifecycle of products is extended. The idea is that the waste of today becomes the raw materials of tomorrow.

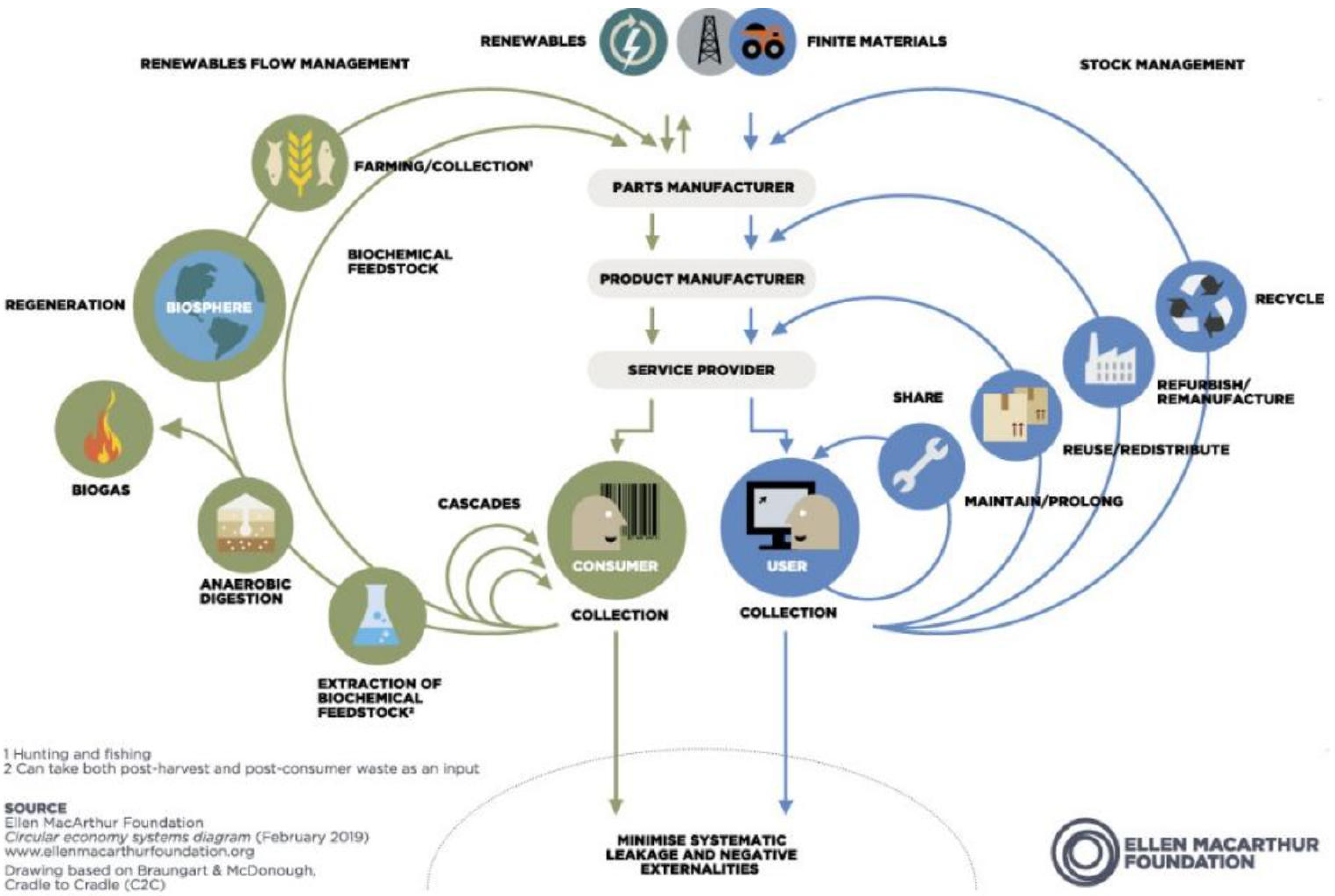
All of these objectives combine to conserve and sustainably manage natural resources including water, soil, forests, wildlife and natural habitats.

Social

Social sustainability centres on the wellbeing of people and the communities they live in. Systems and policies should be directed towards tackling:

- poverty and inequality
- discrimination, prejudice and social exclusion
- lack of access to resources
- insecurity at all scales, from local to global
- poor governance.

The overall objective of social sustainability is to ensure equitable access to opportunities and resources for all people.



▲ **Figure 8.17** The circular economy

Economic

Economic sustainability involves conducting economic activities in a way that preserves and promotes long-term economic wellbeing. Economies become more sustainable by:

» transitioning away from fossil fuels and generating as much energy as possible from renewable sources
» adopting policies and regulations that maximise energy efficiency
» promoting economic models based on the circular economy that reduce waste and limit resource exploitation
» encouraging international cooperation and partnerships between public administration and private enterprises
» ensuring corporate responsibility.

Activities

1 **a** Define environmental sustainability.
  b What are the main objectives of environmental sustainability?
2 Explain the concept of the circular economy.
3 Discuss the difference between social sustainability and economic sustainability.

The techniques used by selected sectors that promote sustainability

Agriculture

The range of management strategies that can be employed to reduce soil degradation in agriculture, including mechanical methods and cropping techniques, can be found in Chapter 10.3 on page 264.

Rock and mineral extraction

Quarrying and mining represent a major human impact on the landscape. However, a range of sustainable practices can reduce the environmental impact of this essential industry. For example:

» Before extraction begins: If there is a reasonable choice of locations, selecting sites strategically close to major transport hubs is an important step in reducing the environmental impact.
» During the active extraction time period: Advanced quarrying methods such as water jetting and laser cutting are more accurate than traditional techniques. They minimise material wastage, produce lower levels of dust and do not involve the use of hazardous chemicals or explosives.
» After the quarry has closed: Phased restoration has become more mainstream in recent decades. This

involves returning parts of the quarried area to a more natural state as quickly as possible after operations have ceased on a part of the site. Such a strategy allows ecosystems to begin their recovery process sooner.

Urban areas

Sustainable practices in combating urban pollution have become more commonplace globally in recent decades. The health and other costs associated with doing the bare minimum have become all too obvious in most cities. A range of techniques can be used to reduce air pollution from vehicles:

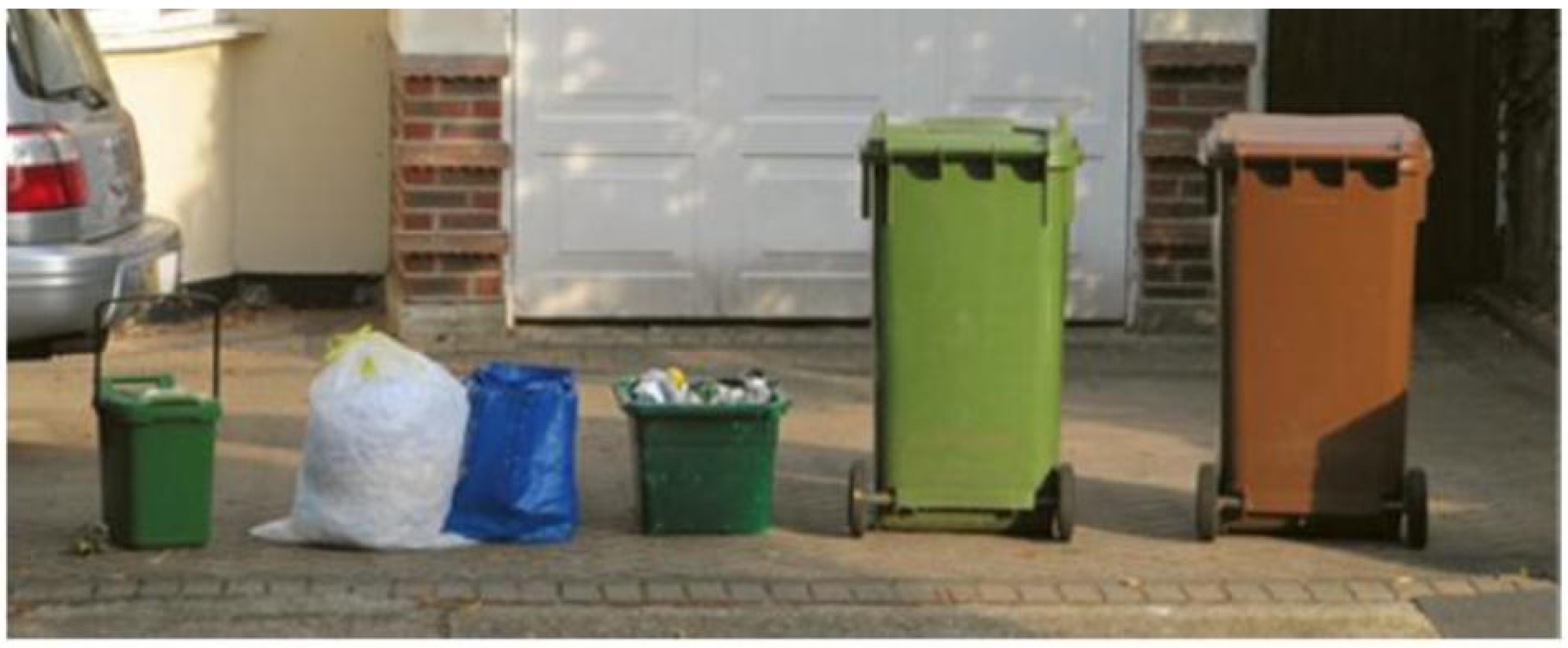
- In the UK, the Electric Car Scheme is a government tax incentive to make electric cars more financially accessible for the average person. EV charging stations are appearing in many sites across the UK.
- In Taipei (Taiwan), smart traffic lights form part of an intelligent adaptive traffic signal system. The objective is smooth traffic flow to reduce the delays and engine idling that contribute to pollution.
- In Copenhagen (Denmark), the Green Wave technology initiative coordinates traffic lights for cyclists, encouraging the use of cycling as an alternative to driving.
- In London, the **Ultra Low Emission Zone (ULEZ)** has significantly reduced the number of older, high-polluting vehicles entering the city (Figure 8.18). ULEZ was introduced in 2019 and initially confined to Central London. It was extended to broadly include inner London in 2021, and then further extended to cover all of Greater London in 2023. The scheme is claimed to be the largest clean air zone in the world.

▲ **Figure 8.18** London's Ultra Low Emission Zone, introduced in 2019

Sustainable techniques introduced to improve the quality of life in other aspects of urban living include the following:

- Urban gardens: Many cities have tried to increase their areas of green space. Plants absorb pollutants and release oxygen, helping to improve air quality. Greater tree and ground vegetation coverage can ease cooling needs and reduce the use of air conditioning, which is very energy intensive. Higher vegetation cover can help mitigate the urban heat island effect.
- Domestic combustion: Smoke Control Areas (SCAs) are in place across the UK, with similar schemes in other countries. This legislation places restrictions on fuels used and smoke released by households. Gas boilers, which are a significant source of nitrogen oxides (NOx) emissions, are coming under close scrutiny, with encouragement to change to alternatives such as electric boilers and heat pumps.
- Construction: The construction industry is undertaking sustainable changes in the way that it operates in construction sites and in the design and operation of the buildings themselves. Sustainability measures on construction sites include:
  - implementing sustainable design, engineering and construction practices to reduce emissions throughout a project's lifecycle
  - using logistics processes that optimise deliveries to reduce mileage, emissions and carbon footprint
  - operating equipment in an energy-efficient manner.

  In terms of construction's end product, finished buildings, the UN set ambitious targets for sustainability at the beginning of this decade. For example, the target reduction rate for energy intensity per m^2 in buildings is 30 per cent by 2030.
- Waste management: At the household scale, people have been encouraged to reduce, reuse and recycle (Figures 8.19 and 8.20). Minimising and repurposing help to preserve the environment. For those with gardens, composting can reduce land pollution. At the municipal scale, **sanitary landfills** are being increasingly used in and around urban areas for land disposal of solid waste. These facilities are designed to control leachate and methane, thus minimising the risk of land pollution from solid waste disposal. Sanitary landfill sites are prepared with impermeable bottom liners to collect leachate and prevent contamination of groundwater.



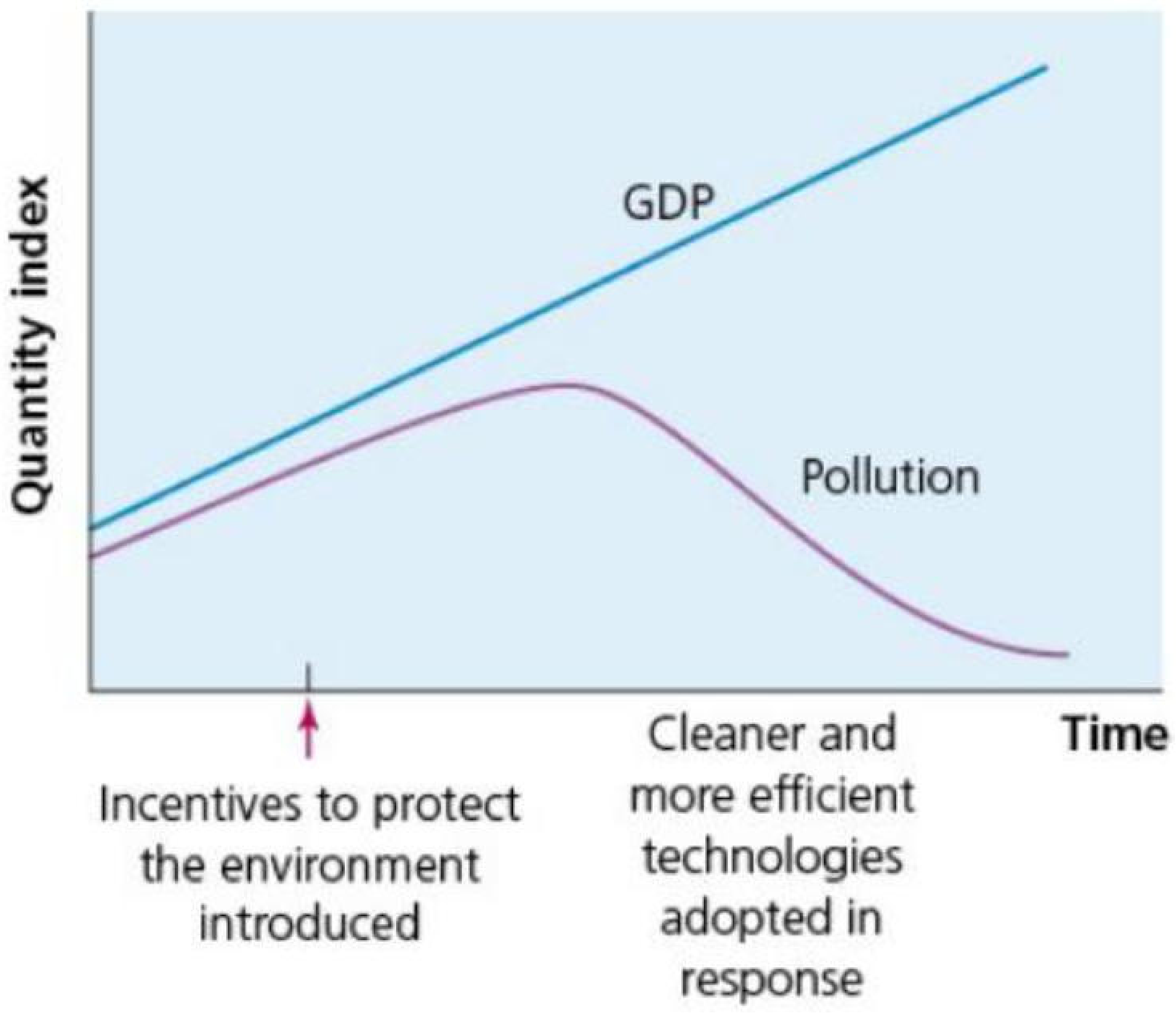
▲ **Figure 8.19** The various categories of recycling

▲ **Figure 8.20** Green waste recycling in the UK

Industry

In developed countries in particular, industrial emissions have reduced substantially since the 1990s due to a combination of stricter emission standards and fuel changes to greener energy. Specific changes include the increasing use of electric boilers, electric arc furnaces and induction furnaces, and large-scale industrial heat pumps.

Industry has spent increasing amounts on research and development to reduce its environmental impact – the so called greening of industry. In general, after a certain stage of economic development, the level of pollution will decline (Figure 8.21). This is because countries have become more aware of their environmental problems, and they have also created the wealth to invest in improving the environment.



▲ **Figure 8.21** The relationship between GDP and pollution

Tourism

A consideration of sustainable development relating to tourism can be found in Topic 9 on page 241.

Activities

1 Explain the benefits of three sustainable techniques used in agriculture.
2 Comment on the examples given on page 195 to improve sustainability in rock and mineral extraction.
3 What are sanitary landfills and why are they important?

The strategies to reduce uneven development

The development gap is such a major global issue that it is not surprising there are different views held by governments, organisations and individuals. For example:

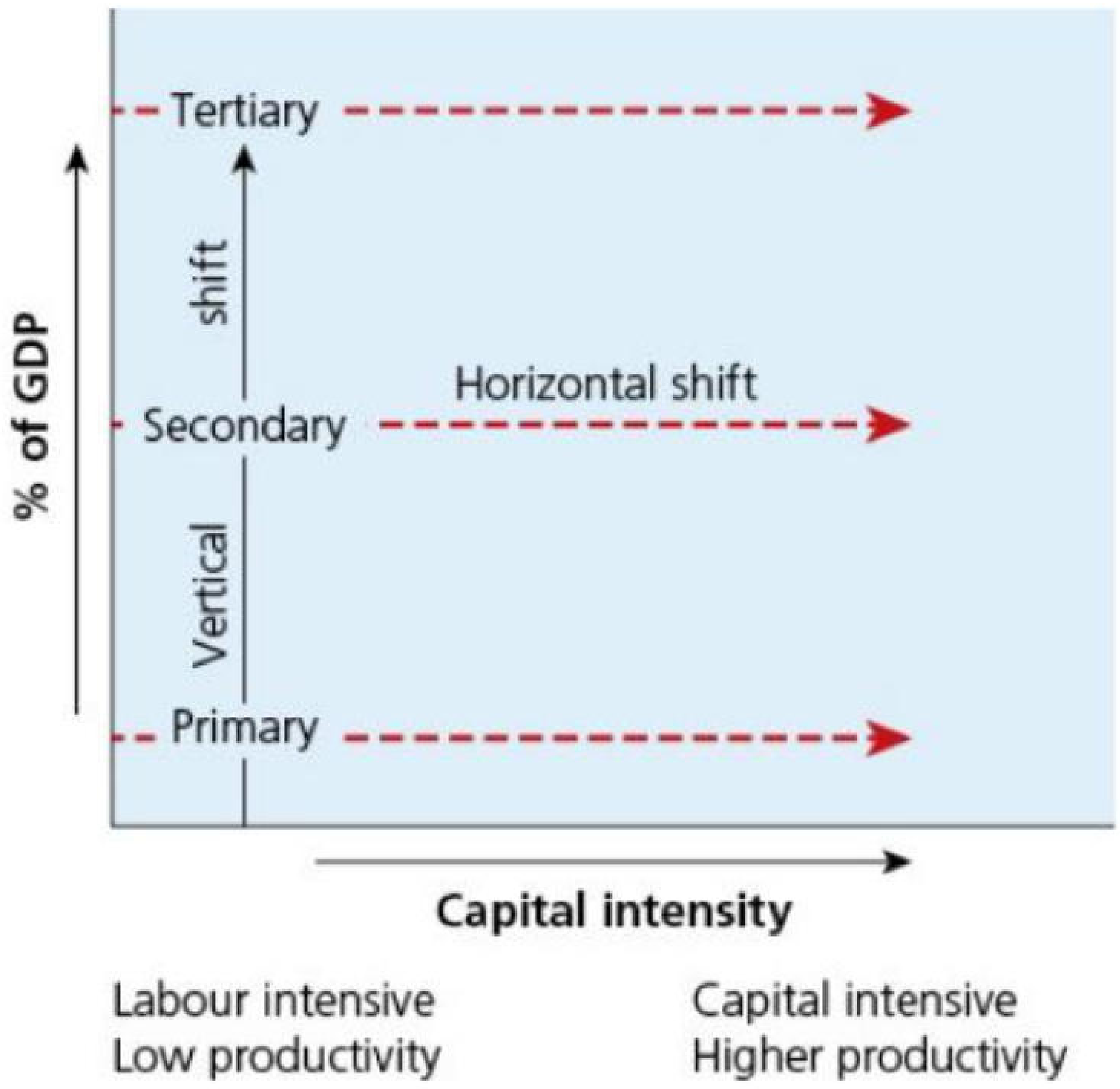
» Some countries contribute a much higher proportion of their GDP to international development than others.
» The political character of a government can influence its attitude to international development.
» Development economists can differ in their views about the best ways to encourage development.
» The strength of **global civil society** in a country can be a big factor in shaping development policy.
» The respective benefits of prioritising trade or aid have been debated for a long time.

Trade

The **United Nations Conference on Trade and Development (UNCTAD)** was established in 1964 to promote the interests of developing countries in world trade. UNCTAD recently stated, 'Globalisation, including a phenomenal expansion of trade, has helped lift millions out of poverty. But not nearly enough people have benefited. And tremendous challenges remain.' There is a strong relationship between trade and development. UNCTAD has stated that even small changes in agricultural employment opportunities or farm prices can have major socioeconomic effects in developing countries.

UNCTAD's Trade and Development Report 2021 highlighted the development potential of:

» the 'vertical shift' in production from primary to secondary, and on to higher-end tertiary GDP and employment (Figure 8.22)
» the 'horizontal shift' of resources from lower to higher productivity. Here all sectors of the economy benefit from becoming more capital intensive and thus less labour intensive.



▲ **Figure 8.22** Sector and productivity change

The Report stated, 'Together, these processes have, in almost all successful development experiences, facilitated a more diversified structure of economic activity, raised productivity and led to an improvement across a broad set of social indicators, including poverty reduction.' More diversified economies are less vulnerable to external shocks.

Reforming the World Trade Organization

The World Trade Organization (WTO) deals with the rules of world trade. Its main aim is to ensure that world trade flows as freely as possible. The WTO was set up in 1995 with far greater powers than its predecessor (the General Agreement on Tariffs and Trade – GATT) to settle trade disputes. Its headquarters is in Geneva, Switzerland. The WTO has over 150 member countries, accounting for more than 97 per cent of world trade.

Today, average tariffs are only about a tenth of what they were in 1947 when the GATT was formed, and world trade has been increasing at a faster rate than GDP. However, in some areas protectionism is still an issue, particularly in the sectors of clothing, textiles and agriculture. Although most countries accept the principle of reducing trade barriers, they may worry that important domestic industries may not be able to compete with lower-cost goods being imported from abroad. As a result, trade agreements can often be very difficult to broker, and can take a long time to achieve.

Critics of the WTO complain it is dominated by the largest economies in the world, with little regard for the trade difficulties faced by many less affluent countries. Organisations such as the Global Policy Forum argue that world trade is very unequal, noting that:

» agricultural subsidies and trade barriers in the USA, EU and other HICs prevent LICs from gaining proper access to the most important markets in the world
» at the same time, LICs have been expected to open up their markets to competition from more advanced nations
» trade is heavily dominated by transnational corporations (TNCs), which often benefit the most from new trade rules
» giving developing countries greater access to markets in developed countries would be a major boost to the incomes of LICs.

Many non-governmental organisations (NGOs) have criticised the ways in which the global trading system operates. Table 8.9 summarises these complaints.

▼ **Table 8.9** Criticisms of the WTO

Dumping	Dumping is the export of products (for instance clothing) at a price less than the normal value of that product. This unfair trade practice can have a big negative impact on developing countries through these imported goods being cheaper than domestically produced goods.
Market access	The conditions attached to IMF–World Bank programmes have often forced developing countries to open their markets, regardless of the impact. Developing countries have often found it difficult to access markets in developed countries such as the EU and the USA.
Patent rights	The high level of international protection on intellectual property rights on essential products such as computer software and medicines has created ‘unnecessary’ costs for developing countries.
Corporate practices	The objective of many TNCs is to produce as cheaply as possible. This often results in the exploitation of people and environments in developing countries.
Uneven influence	Developing countries have long argued that their concerns about the ways in which the global trading system operates are largely overlooked.

Fairtrade

Fairtrade is a system designed to help small-scale producers in developing countries achieve sustainable and fair trade relationships. It recognises the small proportion of the final price of a product that goes to producers. For example, the great majority of the money generated by the tea industry goes to the post-raw-material stages (processing, distributing and retailing), usually benefiting companies in wealthy countries rather than the producers in LICs.

Many supermarkets and other large stores in developed countries now stock some ‘fairly traded’ products (Figure 8.23). Many are agricultural products such as bananas, orange juice, nuts, coffee and tea. However, the market in non-food goods such as textiles and handicrafts has increased in recent years. The Fairtrade system operates as follows:

» Small-scale producers group together to form a cooperative.
» These cooperatives deal directly with companies such as large supermarkets in more affluent countries. Dealing directly cuts out intermediaries, such as wholesalers.
» These companies pay more than the world market price for the products traded.
» The higher price achieved by the cooperatives provides both a better standard of living and some money to invest in the farms of producers.

Advocates of the Fairtrade system say that it is a model of how world trade should be organised to tackle global poverty. They also stress the high environmental standards set by the Fairtrade system. This system of trade began in the 1960s with Dutch consumers supporting Nicaraguan farmers. Today more than 1.9 million farmers and workers are in Fairtrade-certified producer organisations – there are 1880 of these organisations in 71 countries. Currently, Fairtrade is less than 1 per cent of total world trade. NGOs would like to see this system increased in scale.

As you might expect, the Fairtrade system has not been without its problems and critics. For example, some retailers in developed countries have withdrawn from the system in favour of organising their own way of helping producers in LICs.

▲ **Figure 8.23** An example of a popular Fairtrade product

International aid

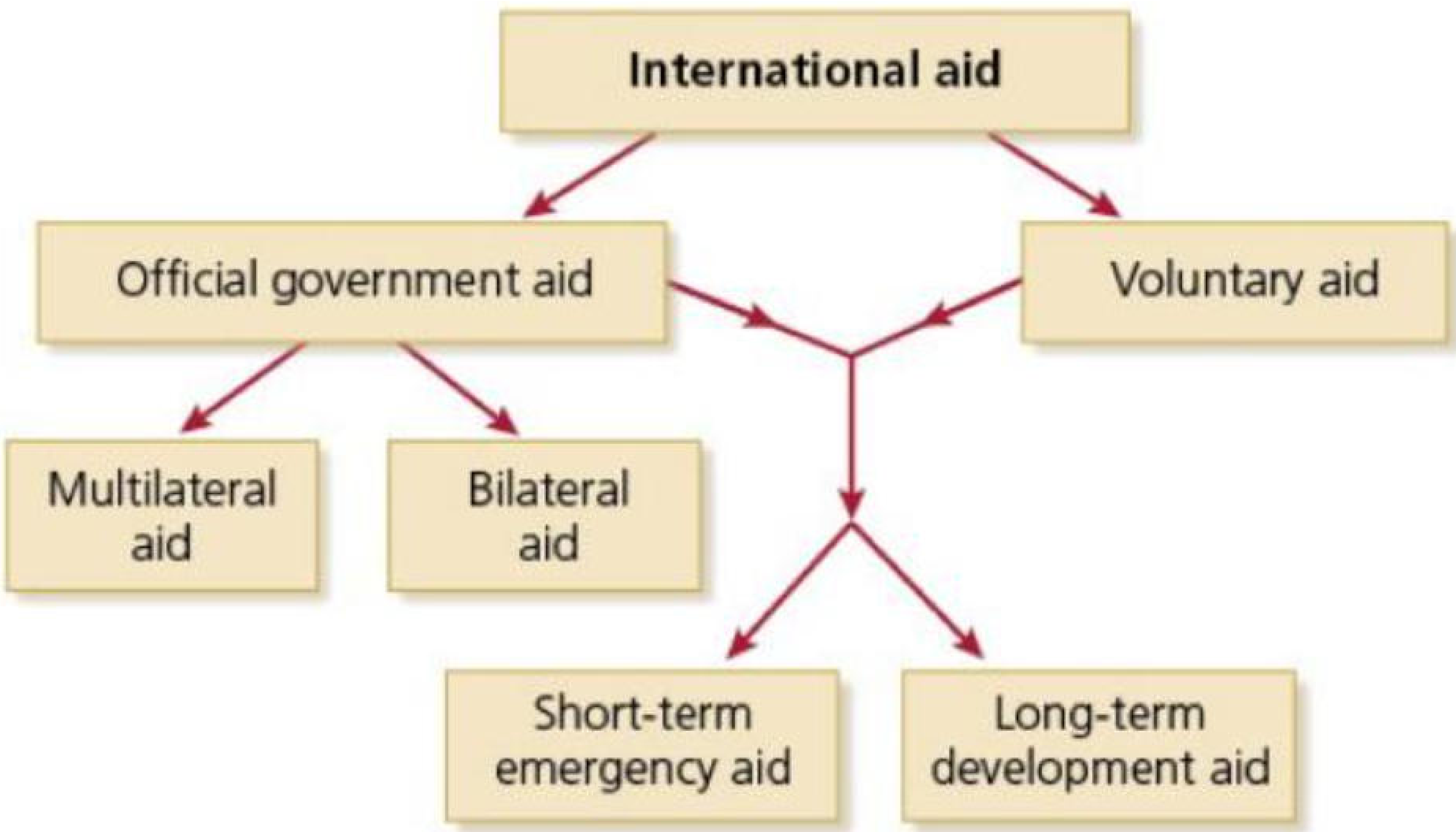
International aid is assistance given to countries in need, in the form of grants or cheap loans. Most developing countries have been keen to accept foreign aid. This is because many:

» lack the hard currency to pay for vital imports
» have insufficient savings to invest in industry and infrastructure
» lack important technical skills.

The origins of international aid date back to the Marshall Plan of the late 1940s. This was when the USA set out to reconstruct the war-torn economies of Western Europe and Japan as a means of containing the international spread of communism. By the mid-1950s the battle for influence between West and East in the developing world began to have a marked effect on the geography of aid. Thus, it is not surprising that reservations are often expressed about the directions of international aid and the reasons behind it.

The basic distinction in international aid is between official government aid and voluntary aid (Figure 8.24):

» Official government aid is where the amount of aid given and who it is given to is decided by the government of the country/countries giving the aid.
» Voluntary aid is run by NGOs or charities, which raise money in various ways from individuals and organisations. However, an increasing amount of government money goes to NGOs because of their special expertise in running aid programmes efficiently.



▲ **Figure 8.24** The different types of international aid

Official government aid can be divided into:

» bilateral aid, which is given directly from one country to another. A significant proportion of bilateral aid is 'tied', despite continual efforts to end this process
» multilateral aid, which is provided by many countries and organised by an international body such as the UN.

Aid supplied to lower-income countries is of two types:

» Short-term emergency aid, often called 'relief aid', is provided to help countries cope with unexpected disasters such as earthquakes, volcanic eruptions and tropical cyclones.

» Long-term development aid is directed towards continuous improvement in the quality of life in an LIC.

There is no doubt that many countries have benefited from international aid. All the countries that have developed from low income to middle income have received international aid. However, their development has been due to other reasons too. It is difficult to be precise about the contribution of international aid to the development of each country. The main concerns about aid are that:

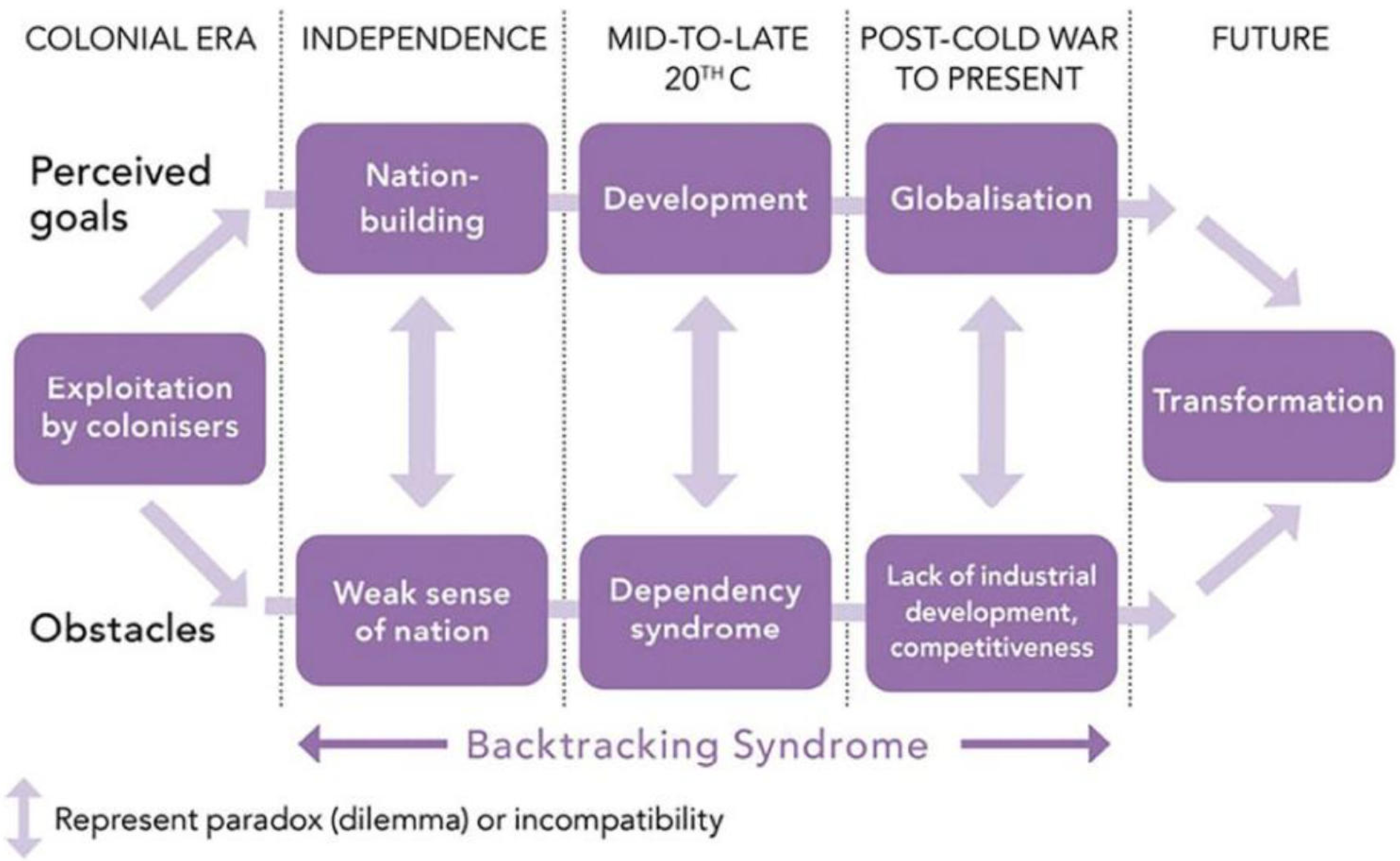
» a significant proportion is **tied aid**
» too often aid fails to reach those that need it most
» the use of aid on large capital-intensive projects may worsen local poverty
» aid may delay the introduction of reforms, for example the substitution of food aid for land reform
» aid can create a culture of dependency.

Figure 8.25 relates to Sub-Saharan Africa, showing the clash between perceived goals and the obstacles to development in the post-colonial era. The author of the diagram, Dr Jong-Dae Park, argues that aid, although well-intentioned, created a culture of dependency in many countries in Africa. Many argue that this state of dependency was imposed on low-income nations as an expression of neo-colonialism. As a result, the hoped-for pace of development was rarely achieved. Dr Park notes:

» Humanitarian aid/disaster relief tends to be successful where donors and recipients have very easy and clear-cut relationships.
» Development aid tends to have problems where relationships between donor and recipient are far more complicated.

He adds that cultural differences and misunderstandings can cause considerable difficulties in achieving goals, particularly with large-scale projects.

Historically, India has been the biggest recipient of foreign aid. This has been because of the country's size and the scale of its development problems. However, such aid has declined rapidly in recent years as the country has developed. India itself now sends aid to other countries such as Bhutan, Nepal, the Maldives, Sri Lanka and Afghanistan. India is not the only emerging country to change from being a recipient of aid to an aid donor. Some advocates of international aid say this shows how successful aid can be in the development process. Nonetheless, the challenges for development will vary between countries and this route may not work for all nations.



▲ **Figure 8.25** The post-colonial period in Sub-Saharan Africa

Many development economists argue that there are two issues more important to development than aid:

» Changing the terms of trade so that low-income countries get a fairer share of the benefits of world trade.
» Writing off the debts of LICs, which were exploitative and impossible to pay back because they were at such a high level.

NGOs have often been much better than government agencies at directing aid towards sustainable development. The selective nature of such aid has targeted the poorest communities, using **intermediate technology** and involving local people in decision-making.

Debt relief

Many development organisations have singled out **debt** as the main problem for the world's LICs. The term 'debt' generally refers to foreign debt. This is money owed to creditors outside the country. This debt has two components:

- Debt outstanding, which is the total amount owed.
- Debt service, which is the interest to be paid each year.

Some of these debts were imposed as 'payback' for what Western countries perceived they had lost when their colonies gained independence. Other loans were set at levels of interest that were effectively unserviceable. Debt affects a country's credit rating and thus its overall economic vulnerability. The **debt service ratio** of many LICs is at a very high level compared to their ability to pay. The debt service ratio is the proportion of a country's export earnings that it needs to use to meet its debt repayments.

Restructuring the debt of LICs began in a limited way in the 1950s. The Heavily Indebted Poor Countries (HIPC) Initiative was established in 1996 by the IMF and World Bank. In 2006, the Multilateral Debt Relief Initiative (MDRI) was launched to provide extra support to HIPCs to reach the Millennium Development Goals. **Debt relief** frees up resources for social spending.

Debt relief is part of a much larger process that includes international aid, which is designed to address the development needs of LICs. Unfortunately, indebtedness grew across many regions from the start of the Covid-19 pandemic. Debt burdens are too high and export revenues too low in many parts of the developing world.

Microcredit and social business

The development of the Grameen Bank in Bangladesh illustrates the power of **microcredit** in the battle against poverty. The idea of microcredit came from the economist Muhammad Yunus, who opened the Grameen Bank in 1983. The Grameen Foundation advocates microfinance and innovative technology to fight global poverty and bring opportunities to those in extreme poverty. The bank provides tiny loans and financial services to these people to start their own businesses. Women are the beneficiaries of most of these loans. A typical loan might be used to buy a cow in order to sell milk to fellow villagers, or to purchase a piece of machinery that can be hired out to other people in the community. The concept has spread well beyond Bangladesh to reach millions of families in over 25 countries (Figure 8.26).

In his book, *Creating a World Without Poverty,* Yunus highlighted '**social business**' as the next phase in the battle against poverty. He presents a vision of a new business model that combines the operation of the free market with the quest for a more humane world.

However, microcredit has faced increasing criticism in recent years because:

- interest rates are often very high
- recipients often use loans to finance consumption, meaning they do not generate any new income that they can use to repay their loan.

▲ **Figure 8.26** Kathmandu, Nepal – financial cooperatives are very important for small businesses to develop

Activities

1. Explain Figure 8.22 (page 197).
2. State one reason why most LICs have been keen to accept foreign aid.

3 What are the two components of debt?
4 List three characteristics of the Fairtrade system.
5 What are the advantages and disadvantages of microcredit?

Detailed specific example

Indonesia: an emerging economy

Indonesia's current level of economic development

According to the World Bank's income classification, Indonesia is an upper-middle-income country. Indonesia's significant economic indicators include:

- an annual GDP of over US$1 trillion
- an increasing, economically active population, with a national median age of 29
- a large and growing middle-class consumer base
- expanding commodity exports
- policies aiming to attract foreign direct investment
- a new capital city under construction in Kalimantan.

Geographical location

Indonesia is the world's largest **archipelagic country** (Figure 8.27), lying between the Indian Ocean and the Pacific Ocean and consisting of over 17,000 islands (Figure 8.28). It lies across the equator, spanning a distance of one-eighth of the Earth's circumference. Indonesia is strategically located along major shipping lanes connecting East Asia, South Asia and Oceania. The country is characterised by active volcanoes and is bounded to the south and west by a series of deep-ocean trenches.

▲ **Figure 8.27** Indonesia lies between the Indian and Pacific Oceans and consists of over 17,000 islands

▲ **Figure 8.28** A small, uninhabited island in the Ceram Sea, Indonesia

The capital city, Jakarta, is located near the northwestern coast of Java. About 70 per cent of the country's population live in coastal areas, relying on the surrounding seas for both income and nutrition. Indonesia contains:

- the third-largest tropical rainforest in the world
- the world's largest tropical peatlands
- the world's largest mangrove forests.

All of these important physical environments store huge amounts of carbon that help mitigate climate change impacts.

Population and economic status

With a population of almost 279 million (Table 8.10), Indonesia is the fourth most populous country in the world after India, China and the USA. Indonesia has the youngest population in the Southeast Asia region, with 24 per cent of its population under 15 years of age. However, the country's rate of population growth has fallen considerably in recent decades. With a birth rate of 17/1000 in 2023, Indonesia is now in stage 3 of demographic transition. Fifty-six per cent of Indonesia's population live on the island of Java, especially in the capital city, Jakarta. Java is the economic core of the country, accounting for 58 per cent of national GDP. Indonesia has the largest Muslim population in the world. It is a democratic country that has endured periods of autocracy in the past.

▼ **Table 8.10** Indonesia fact file for 2023

Population	278.9 million
Birth rate	17/1000
Death rate	6/1000
Infant mortality rate	15/1000
Life expectancy at birth	75 years
Population living in urban areas	59 per cent
Urban population in informal settlements	19 per cent
Population per km^2 of arable land	1060
GNI per capita ($PPP)	14,250
Human Development Index category	High, 112th in the world

In economic terms, Indonesia is the tenth-largest economy in the world by purchasing power parity ($PPP). By 2028, it is projected to be the world's sixth-largest economy by $PPP (Figure 8.29). Indonesia is the largest economy in Southeast Asia and is generally viewed as an emerging economy. The country was invited to join the BRICS (Brazil, Russia, India, China and South Africa) group of countries in 2023, but declined the offer. However, it has not ruled out joining in the future. Indonesia is a member of G20, the group of the world's twenty largest economies. It assumed the rotating G20 Presidency in 2022.

The poverty rate in Indonesia has fallen by more than half since 1999, to under 10 per cent in 2023.

Raising Indonesia's level of development

The government of Indonesia has set a goal for the country to achieve 'high-income' status by 2045. This is consistent with the government's strategy in recent decades to raise the country's levels of development, quality of life and standard of living. The country has been following a 20-year development plan since 2005 (2005–25), which is segmented into five-year Medium-Term National Development Plans (RPJMN), each with different development priorities.

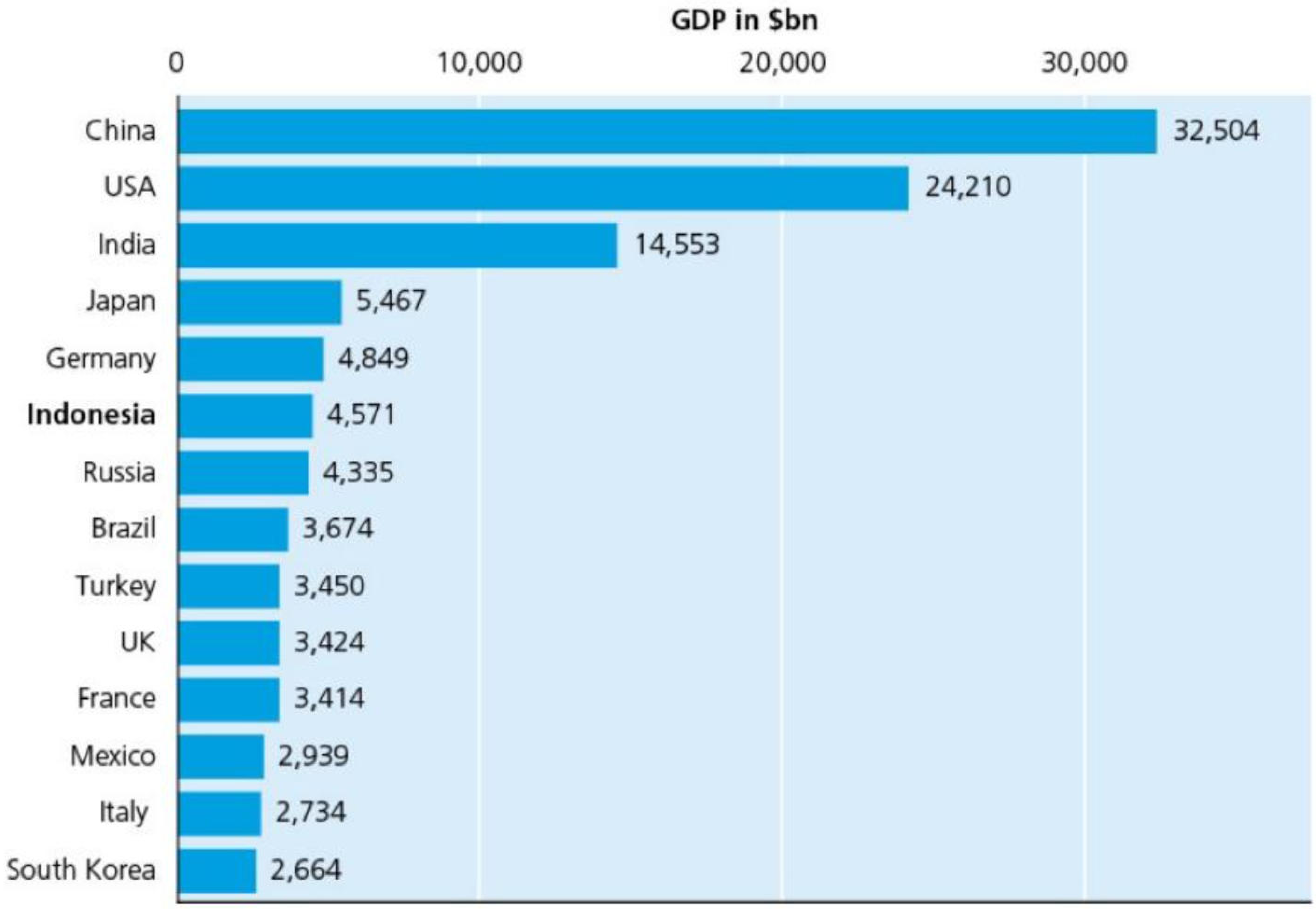
The 2023 Human Development Index ranked Indonesia as a country of medium human development. The quality of health in Indonesia has risen with the government focusing on improving the general health of people in poverty through the use of Universal Health Coverage [UHC] Policies. Examples of other significant improvements in the quality of life have been in the areas of economic stability, food security and the gender gap.

Economic transformation

In the 1980s and 1990s, Indonesia experienced rapid industrialisation through growth in manufacturing exports. This went hand-in-hand with a high rate of urbanisation. Millions of workers moved from agricultural to non-agricultural jobs, particularly in the expanding manufacturing sector. This 'first wave' of **structural transformation** substantially increased average incomes, resulting in a large reduction in the number of people below the poverty line.

However, the nature of structural transformation changed as industrialisation stalled after the 1997–98 Asian financial crisis. Indonesia's economy shifted into a natural resource-based growth model. Although workers continued to move away from agricultural employment in this 'second wave' of structural transformation, they mostly found employment in low-paid service sectors, rather than in better-paid manufacturing jobs. While low-paying jobs in the service sector are generally more productive than agricultural activities, the difference is much less than between agricultural and manufacturing employment.

The perceived lack of middle-income jobs in Indonesia today is mainly due to the more limited productivity gains in the second wave of structural transformation. In this latter period, most of the improvement in labour productivity occurred due to efficiency gains within economic sectors (agriculture, manufacturing, services) rather than the movement between sectors.



▲ **Figure 8.29** The projected largest economies in the world in 2028 (GDP in $bn)

Between 2014 and 2023, Indonesia's economy maintained a growth rate of about 5 per cent (except during the Covid-19 pandemic), with GDP per capita rising to about US$5000 in 2023 (Figure 8.30). However, despite the significant decline in the incidence of poverty, data from the World Bank suggests that 53 per cent of Indonesia's population in 2018 either lived in poverty or were 'economically vulnerable'. In comparison, neighbouring Malaysia had an economically vulnerable population of only 3.7 per cent in 2018. While income inequality remained relatively stable during the first wave of economic transformation, it increased considerably during the second wave.

Economic partners

Indonesia's main trading partners are mainland China, the USA, Japan, Singapore, South Korea and Hong Kong. The country has longstanding trade and investment relationships with Japan, Singapore and South Korea.

In 2022, Indonesia's exports were valued at US$320 billion, up from US$190 billion in 2017. The main export destinations were China (21.2 per cent), the USA (9.9 per cent), Japan (8.3 per cent) and India (7.9 per cent). The main products exported were coal briquettes (15.9 per cent), palm oil (9 per cent), ferroalloys (4.3 per cent), petroleum gas (3.7 per cent), copper ore (2.8 per cent) and lignite (2.6 per cent).

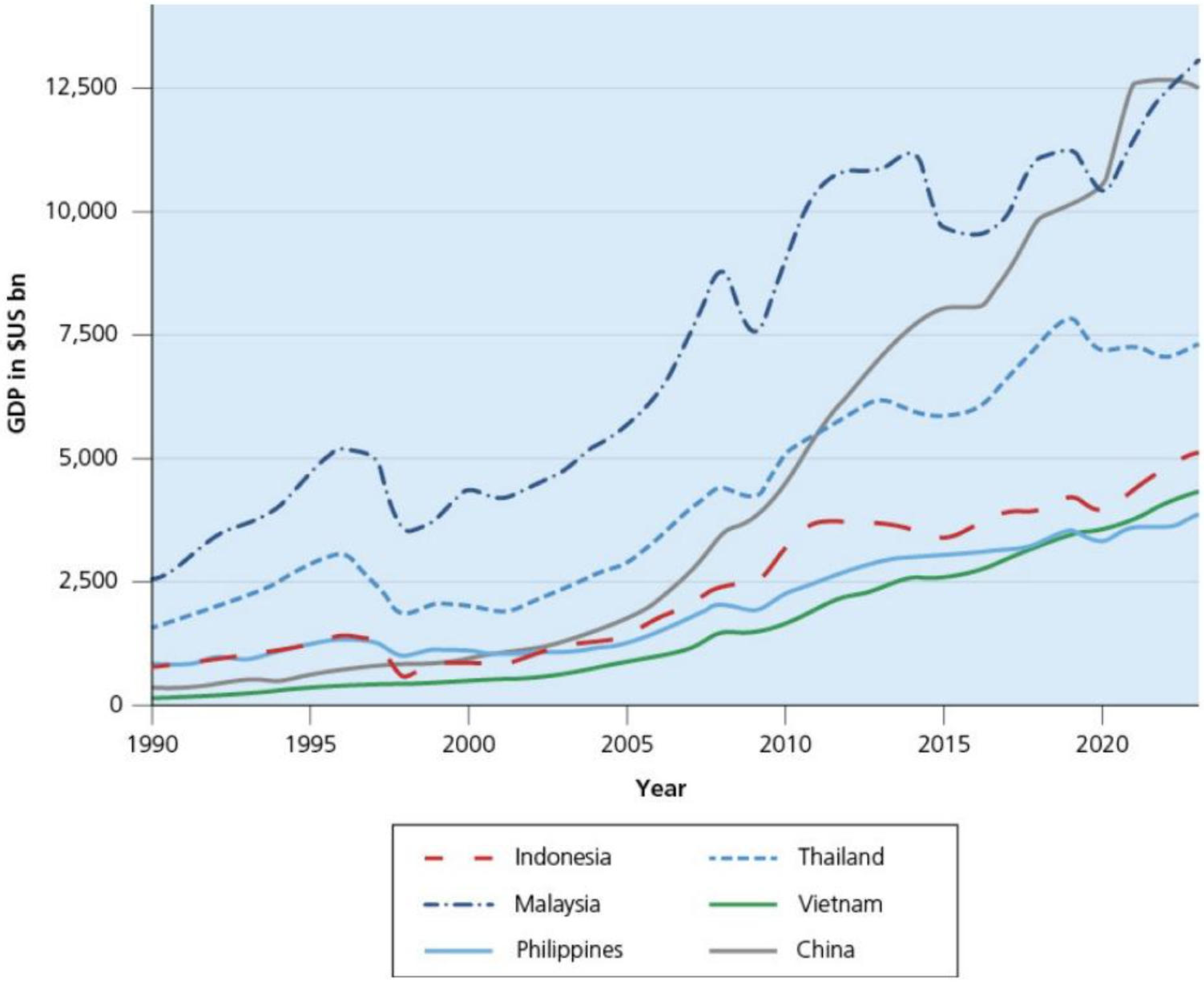
In the same year, imports were valued at US$250 billion. The leading sources of imports were China (28.5 per cent), Singapore (9 per cent), Japan (6 per cent), Malaysia (5 per cent) and Thailand (4.2 per cent). The main products imported were refined petroleum (9.3 per cent), crude petroleum (4 per cent), motor vehicles, parts and accessories (1.7 per cent), and broadcasting equipment (1.6 per cent).

Indonesia had a current account surplus in 2021 for the first time in a decade, and this increased in 2022. However, maintaining a surplus could become a challenge if commodity prices decline.

Focus on digitisation and sustainable industries

Indonesia's digital economy grew from US$41 billion in 2019 to US$77 billion in 2022, with strategies to increase electronic transactions and digital banking activity. A key element of this strategy was the One Data Initiative, a drive to digitise government processes to create more efficient, secure and interconnected data flows between departments. The objectives of this strategy were to:

- increase economic growth through improvements in efficiency
- give potential foreign investors more confidence to begin or expand their operations in Indonesia.



▲ **Figure 8.30** GDP per capita for Indonesia and five other Asian countries, 1990–2023

Digitisation is seen by the government as an important factor in the increase in the number of **unicorns** (start-ups), which is valued at least US$1 billion in the country's fintech, e-commerce and logistics sectors. Fintech is a portmanteau (combination) of the words 'financial' and 'technology'. It describes the use of technology to deliver financial services and products to consumers.

Indonesia has prioritised electric vehicle (EV) battery production as a key industrial sector. The country has the advantage of the world's largest nickel reserves and the third-largest cobalt reserves. Both of these raw materials are essential in the production of batteries for electric vehicles. Indonesia is shifting away from exporting these raw materials, to producing EV batteries domestically, aiming to diversify its economy and attract further foreign direct investment. This policy also forms part of its net-zero strategy. This is an element of a wider focus to attract value-adding industries to Indonesia's large reserves of natural resources. This trend broadens the country's manufacturing export base and adds resilience to its current account balance.

In 2020, Indonesia banned the export of nickel ore, which served to encourage companies such as China's Tsingshan, South Korea's LG and Brazil's Vale to set up more factories in the country in order to access its raw materials. This involves not just more nickel being refined in Indonesia, but also foreign companies building more of their supply chains in the country. Indonesia defied a ruling by the World Trade Organization that the ban was unfair. The country has also introduced export controls on bauxite, the ore used to produce aluminium, and plans to introduce export controls on copper.

In 2023, foreign investment grew 13.7 per cent to reach a record high of US$47.3 billion, after increasing by 46.6 per cent in 2022. About a third of FDI, predominantly from China, has gone into the metals and mining industries since 2021.

Developing infrastructure

The current administration has spent more on infrastructure than any previous government. High levels of investment have gone into roads, railways, dams and ports across the country. The money allocated to infrastructure rose from US$9.9 billion in 2014 to US$27 billion in 2024. Between 2014 and 2024, Indonesia built 16 new airports, 18 new sea ports, 36 new dams and 2100 km of toll roads. Lower logistics and distribution costs are an important incentive for foreign direct investment. However, critics argue that state-owned companies have built up substantial debts with this increase in expenditure on infrastructure.

Geopolitics

Indonesia has been a long-term member of the international Non-Aligned Movement (NAM). Member countries try to maintain good relations with both the 'Western' economies led by the USA, and what has become known as the 'Global

South' under the 'leadership' of China. Indonesia's non-aligned stance is at least part of the reason that it turned down the invitation to join the BRICS in 2023.

Capital city: from Jakarta to Nusantara

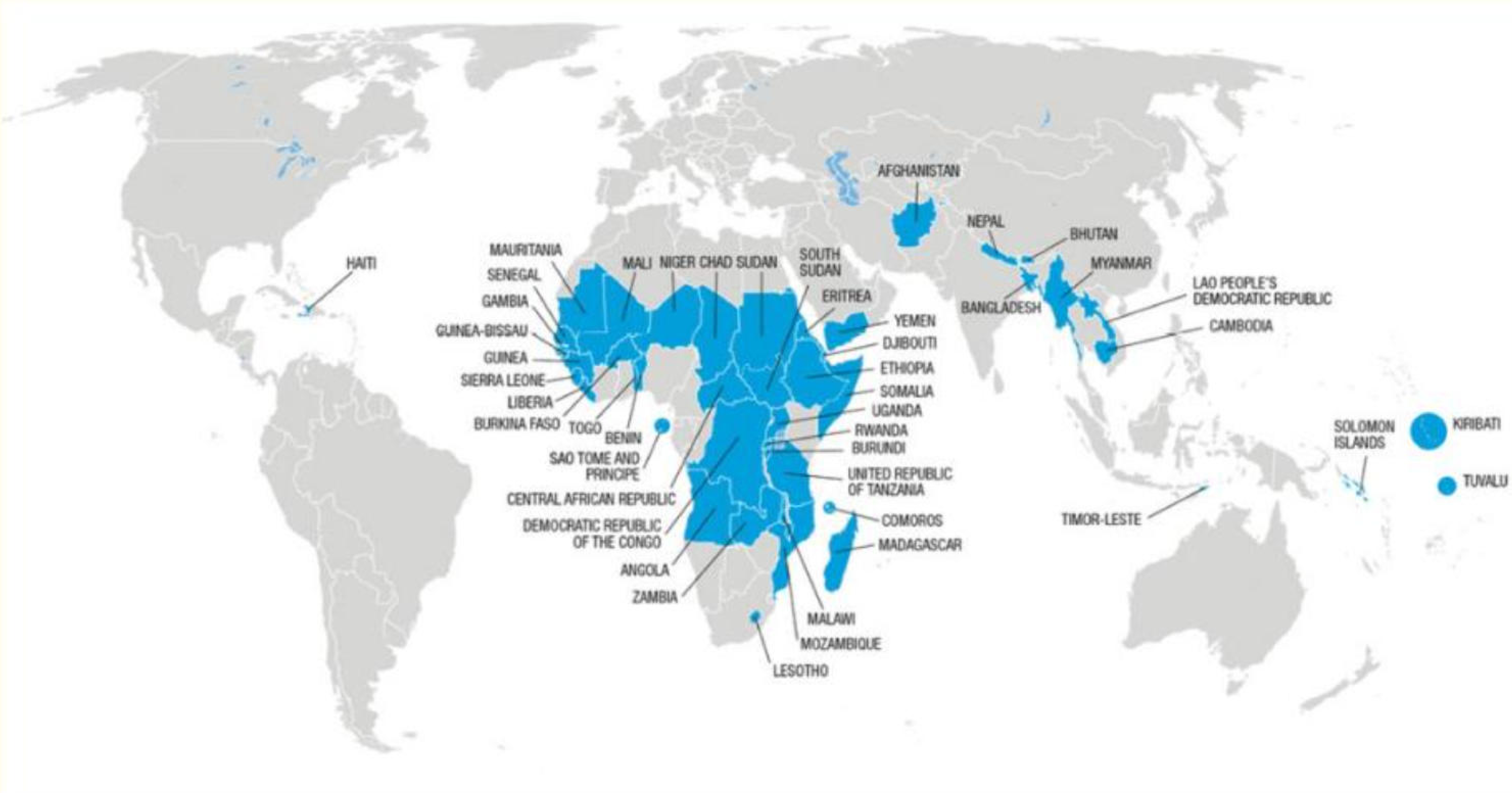
A new capital city called Nusantara (which means *archipelago* in Indonesian) is in the early stages of construction in the heart of the rainforest in Kalimantan (Borneo). This is a US$45 billion project, announced in 2019, to move the government 1236 km away from the overcrowded and subsiding megacity of Jakarta. Jakarta's metropolitan area is home to 30 million people. Construction of the new city began in July 2022 in an area of forests and oil palm plantations, 30 km inland from the Makassar Strait. The first stage of development involves constructing government facilities and other buildings for the expected initial population of 500,000. Construction is planned to be completed by 2045.

Activities

1 What is Indonesia's current level of economic development?
2 Suggest why Indonesia's relatively young population structure is viewed as an economic advantage.
3 Describe Indonesia's two waves of structural transformation.
4 Why is Indonesia investing heavily in digitisation and sustainable industries?
5 Describe and explain the importance for Indonesia of investing in infrastructure.

Practice questions

1 Figure 8.31 shows the world's least-developed countries.
  a Identify the least developed country in the Americas. [1]
  b Describe the distribution of the least developed countries. [3]
  c Suggest why islands such as Kiribati and Tuvalu are among the least developed countries. [3]



▲ **Figure 8.31** The world's least-developed countries

2 Table 8.11 shows life expectancy (years) for selected countries.
  a Calculate the difference in life expectancy between the country with the highest life expectancy and the country with the lowest life expectancy. [2]
  b Compare the characteristics of countries with high life expectancy with those with low life expectancy. [4]

▼ **Table 8.11** Life expectancy (years) for selected countries

Highest life expectancy (2020–25)	Lowest life expectancy (2020–25)
Monaco 89.4	Central African Republic 54.4
Hong Kong 85.3	Chad 55.2
Japan 85.0	Lesotho 55.6
Macau 84.7	Nigeria 55.8
Switzerland 84.3	Sierra Leone 55.9

3 a Identify two economic indicators of development. [2]
  b Suggest how improving levels of literacy may raise levels of development. [3]
4 Using named examples, evaluate ways in which uneven development may be reduced. [7]